A Strategic Environment:

Community Conservation, Site Selection, and Spillovers in Eastern Zambia

Preliminary draft

The latest version of this paper is available at this link.

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Abstract

I evaluate the environmental impacts of a USAID-funded conservation project in Eastern Zambia that established Conservation Areas within Participating Chiefdom boundaries. Using satellite and administrative data at the 1 km × 1 km grid level from 2001 to 2023, I employ a Difference-in-Differences design with propensity-score-matched controls to estimate effects on deforestation across two spatial margins: (i) Conservation Areas and Communal Lands within Participating Chiefdoms, and (ii) areas close to and far from conservation boundaries within Non-Participating Chiefdoms. For Participating Chiefdoms, I found that tree cover loss fell by 0.14–0.16 percentage points within Conservation Areas, with no evidence of deforestation leakage into adjacent Communal Lands. For Non-Participating Chiefdoms, areas close to conservation boundaries exhibit spillover reductions of 0.17–0.19 percentage points in tree cover, and no effects for far areas. In hectares, the sum of these effects corresponds to yearly avoided deforestation of 753-866 ha and 6,777-7,794 ha from 2015 to 2023. Heterogeneity analysis reveals that conservation effects were stronger in areas closer to roads and human settlements. I also document spatial selection bias by showing that Conservation Areas were disproportionately placed in remote, low-deforestation locations, indicating that communities selected areas where the opportunity cost of conservation is lowest. These findings suggest that the aggregate impact of community-based conservation programs depends critically on spatial targeting and the extent to which incentives diffuse beyond formally designated boundaries.

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1 Introduction

From 2001 to 2024, there was a total of 520 Mega hectares of tree cover loss globally, 13% decrease relative to 2000 tree cover area, which generated 195 Gigatons of Carbon Dioxide (CO^2) emissions (Global Forest Watch 2026). Moreover, deforestation accounts for 12% - 20% of global greenhouse gas emissions (GHG), composed of pollutants such as Carbon Dioxide (CO^2), contributors to climate and environmental changes. Additionally, scientific literature shows the negative effects of these pollutants on human development outcomes, such as health, mortality, and cognitive development [1]. Besides the negative aspects of deforestation, forest services are important sources of income. In Sub-Saharan Africa, the Agriculture, Forestry, and Other Land Use Sector is responsible for 65% - 70% of the labor force and 30% - 40% of GDP (Nash, Halewood, and Melhem 2013). To reduce the pace of tree cover loss, policymakers must design interventions that incentivize pro-environmental behavior while considering agents forgone income and outside options from forest-related activities (Correa et al. 2020; Jayachandran 2022; Jack et al. 2022; Cisneros et al. 2022). This is especially relevant in regions such as Eastern Zambia, where households are dependent on activities such as charcoal production or land cleaning for subsistence agriculture (Huntington et al. 2016; BioCarbon Partners 2019).

Zambia has the fourth-largest forest in Africa and has increased its deforestation patterns in recent years. The country has the ninth-highest deforestation rate in the world and 11th in per capita terms (Global Forest Watch 2026). Since 2000, Zambia has lost approximately 11% of its tree cover, mainly driven by clearing land for agriculture, wood extraction, and charcoal production (Huntington et al. 2016; BioCarbon Partners 2019). In 2015, USAID implemented the Community Forests Project (CFP) in partnership with BioCarbon Partners focusing on protecting the Luangwa Corridor, an important ecological corridor. The project established contracts with chiefdoms to conserve areas within their boundaries. These zones serve as reference areas for selling carbon offsets and generating revenue to invest in community projects within the participating chiefdoms. These activities aim to implement transition activities to reduce forest-service dependency and spread sustainable production alternatives. These contracts have a validity of 30 years to create long-term relationships with local communities. This is one of the largest REDD+ projects in Africa by land area, mobilizing 14 million dollars and protecting 943,676 hectares (BioCarbon Partners 2019).

1. For more information about the effects of pollutants on human health and outcomes, see Bedi et al. (2021), Fu, Viard, and Zhang (2021), Odo et al. (2023), and Heyes and Saberian (2024)

I use the USAID-funded REDD+ program in Eastern Zambia to study the impacts of financially rewarding communities' conservation on annual tree cover loss. I also investigate how selection bias influences conservation boundaries establishment. The program established conservation areas within the geographical boundaries of participating Chiefdoms, committing to financial rewards for verified avoided deforestation in these zones. Because they were established in communal lands ruled by chiefdom leaders, these boundaries are determined through interactions with local leaders. This local feature is a potential mechanism for selection bias from leaders, who have incentives to choose low conservation cost areas to maximize rewards due to conservation. To investigate site selection bias and tree cover loss effects, I construct a high-resolution panel dataset of 1km × 1km grids spanning from 2001 to 2023 by integrating the program records and geospatial data. I incorporate detailed information from USAID on newly established Protected Areas boundaries and participant chiefdoms. This geographical information is merged with satellite-based measures of tree cover and forest loss from Hansen et al. (2013). To control for possible confounding factors and match cells on geospatial characteristics, I use additional geospatial characteristics such as agricultural land productivity, settlement locations, altitude, distances to roads, water bodies, and the electrical grid. For my heterogeneity analysis, I categorize cells into different groups based on their remoteness and human settlement presence. Finally, to account for long-term political and institutional boundaries, I use historical pre-colonial chiefdom boundaries from Baldwin (2013).

To investigate selection bias on conservation boundaries, I use a logit model to estimate conservation likelihood based on grid characteristics. Using grid conservation status, I identify geospatial and pre-program features that predict grid status after program implementation. To minimize concerns with selection bias, I estimate effects on annual tree cover loss combining grid matching with a Difference-in-Differences (DiD) framework. This strategy allows me to estimate the causal effects of the program by comparing grids similar in the likelihood of conservation and following parallel trends before the program. To address spillover concerns in adjacent regions, the estimation compares grids located in the Luangwa Corridor, an ecologically significant area targeted by the program, with those outside the corridor. To test for heterogeneity, I split the participant chiefdom into conservation and communal land zones to test if differential financial incentives between areas may lead to potential deforestation leakages into communal land. As conservation is rewarded only in conservation zones, agents face incentives to shift forest-related activities to communal land zones, where tree cover loss

is not punished. I further explore potential spillover effects on the non-participant chiefdoms grid by examining deforestation trends in chiefdoms that are in the same region but do not participate in the program. Finally, I analyze heterogeneous effects conditional on grid-level characteristics using measures of grid remoteness and human presence.

Although there is extensive literature on studying different conservation policies' effects, how to design effective conservation policies is an open question as multiple factors influence the effectiveness of these policies (Balboni et al. 2023; Jayachandran 2022). For instance, conservation take-up is a crucial element for policy success, but few papers address this issue (Jayachandran 2022). Previous work shows that institutional dimensions such as trust in contract realization, corruption, and political representation influence program take-up and effectiveness (Burgess et al. 2012; Cisneros and Kis-Katos 2022; Jack et al. 2022; Gulzar, Lal, and Pasquale 2024). Property rights serve as a fundamental institutional prerequisite for effective conservation. By clearly defining the boundaries of responsibility, secure tenure mitigates the problem of asymmetric information, allowing monitoring systems to accurately attribute environmental outcomes to specific agents. The ability to monitor behavior ensures that rewards for preservation and sanctions for degradation are directed toward the appropriate parties, thereby aligning individual incentives with collective conservation goals (Baragwanath and Bayi 2020, Sze et al. 2022; Baland et al. 2010). The amount of financial incentives is an important aspect of these programs. Policies must consider compensations proportional to opportunity costs of agents when pursuing conservation actions such as not deforesting. Land use models incorporate maximization processes in choosing to engage in environmentally destructive activities, such as the benefits of forests for communities or the benefits of land change for agricultural production. The costs of no deforestation can be related to different aspects, such as forest income dependency and necessary skill formation to migrate to non-forest markets (Jayachandran 2022; Izquierdo-Tort, Jayachandran, and Saavedra 2024). Finally, long-term participation is important for the sustainability element of the programs. Conservation projects demand considerable funding to compensate agents for pro-environmental behaviors, and the end of financial support can lead agents to return to forest-related services (Etchart et al. 2020).

I contribute to the literature in the following dimensions. First, I am the first to evaluate this project, which is one of the largest REDD+ projects by land protection areas in the African continent. Second, I add to understanding how financial compensation community programs can incentivize conservation in the context of communal lands (Sims 2010; Jayachandran et

al. 2017; Giudice et al. 2019; Jack et al. 2022; Malan et al. 2024). Communal land is an important setting as this is present in countries where land ownership is related to historical or ethnic origins, such as chiefdoms in African countries or Native land demarcations in Latin America (BenYishay et al. 2017; Giudice et al. 2019; Sze et al. 2022; Rico-Straffon et al. 2023). Third, I add evidence on selection bias in the designation of protected areas in the context of chiefdom-owned lands. The endogenous nature of the establishment of conservation zones allows me to understand how local authorities may strategically allocate areas that are less costly to conserve. Fourth, I provide results on conservation program spillover effects on neighboring areas by estimating the effects on communal land located within participating chiefdoms and adjacent grids within non-participating chiefdoms.

I find selection bias in the decision of conservation zones establishment, which are more likely to be located in grids with lower pre-policy tree cover loss rates and greater remoteness. This finding highlights the importance of accounting for the incentives created by conservation policy design. Because program payments are tied to avoided deforestation within protection boundaries, local decision-makers have an incentive to select areas with already low deforestation pressure to maximize rewards. For the causal effects on tree cover loss, I do not find a significant overall effect at the chiefdom level. However, there is consistent evidence of reduced tree cover loss rate in conservation areas (CAs), suggesting that financially rewarding conservation is effective. The ATE estimates of -0.15 to -0.14 pp reduction correspond to approximately to 88 % deforestation averted in comparison to the control group. Importantly, there is no evidence of deforestation leakage in communal land zones in participant chiefdoms, where there are no financial incentives for conservation. Conservation spillovers are present in non-participant chiefdoms. I find evidence of a reduction in deforestation in grids close to conservation boundaries after the policy implementation. The coefficients of -0.14 to -0.16 pp suggest approximately a 60% reduction in deforestation compared to the comparison group. This effect has a decreasing distance gradient, which shows closer areas with a stronger decrease in tree cover loss. In hectares, the sum of the conservation areas and close area spillover effects corresponds to a yearly avoided deforestation of 753-866 ha and 6,777-7,794 ha from 2015 to 2023.

The grid heterogeneity analysis shows that the largest post-policy deforestation reductions occurred in observations closer to roads and human settlements, suggesting that conservation efforts were particularly effective in regions facing higher deforestation pressure. I

provide additional evidence of parallel trend assumptions and the robustness of my results to different matching algorithms. These results indicate that while the program successfully reduced deforestation within and near conservation areas, its overall impact was constrained by strategic site selection, which favored low-risk areas. These results highlight the importance of designing conservation programs that align incentives with broader environmental goals.

2 Institutional Background

In 2015, USAID launched the Community Forests Program (CFP) in Eastern Zambia to reduce deforestation in the Eastern region of Zambia. The project implementation cost was approximately 14 million US dollars and was run in partnership with BioCarbon Partners, a firm specializing in forest conservation projects. The object of the project is to protect a natural corridor called the Luangwa Valley, where there is an important presence of biodiversity and nature related activities such as safaris and sport hunting. Figure 1 highlights the ecological corridor, mainly concentrated in the eastern region extending to Malawi and South of Zambia. According to BioCarbon Partners (2019), the intervention successfully institutionalized a minimum of 700,000 protected hectares within the valley, reaching 943,676 ha and consolidating as one of the largest REDD+ projects in Africa. The intervention establishes a 30 year long contract relationship with 12 participating chiefdoms that are financially rewarded if they avoid deforestation in established conservation areas. Using estimated carbon storage in these zones, the firm sells carbon offsets to finance infrastructure and mitigation activities in communities within the participating chiefdom.

Households in this region face a lack of access to important services and are socioeconomically vulnerable. According to Huntington et al. (2016) and BioCarbon Partners (2019), 87.4% of the population lives in rural areas and does not have access to electricity, public water or sanitation infrastructure. Subsistence agriculture is an important component of household income, comprising up to 64% of it. Another important source of income is the production of charcoal, commonly used for cooking by households. These populations are also economically vulnerable, with 75% of them living with less than $1.5 per day and 60% living in extreme poverty. These facts highlight the financial constraints of communities and the dependence on forest use for income, household activities, and nutrition, a clear challenge to the success of the program.

Financial contracts are established at the chiefdom level in partnership with chiefs, gov-

ernment agencies, and local authorities. For each chiefdom, the REDD+ conservation areas (Figure 1) were delimited after interactions with local communities and authorities. Through these negotiations, they determine areas that would be assigned to protection and used as reference areas to measure carbon storage. These contracts are valid for 30 years to create a long-term commitment with chiefdoms and maintain conservation efforts. Under the contract, BioCarbon Partners committed to pursue Verified Carbon Standard (VCS) certification and reinvest carbon revenues in community development projects. The VCS verification was completed in October 2019, after the end of the USAID-funded phase, covering the program's first monitoring period (May 2015–December 2018). Verified carbon credits corresponding to this period were issued and sold on voluntary carbon markets. Verification has continued annually thereafter, with the 30-year contract running until 2045. This timeline implies two distinct incentive phases. In the first cycle (2015–2018) communities receive program investments in infrastructure and alternative livelihoods, in addition to USAID-funded conservation flat fees paid conditional on the size of their designated conservation areas in 2015. In the second phase, starting in 2019, carbon revenue cycles provided an additional income stream but now conditional on the preservation of designated areas (BioCarbon Partners 2019). This extended component can potentially lead to a long-term incentive to protect areas, as conservation revenue keeps incentives through reinvestment into communities.

To incentivise conservation in regular communal areas, where conservation is not financially rewarded, the program financed basic infrastructure and alternatives to income. Carbon credit revenues are transferred to chiefdoms that decide the allocation of these resources through local administration boards called Community Resources Board (CRBs) and Village Action Groups (VAGs). The chiefdom authorities then decided how to invest the resources in their communities. However, investments are restricted to the provision of physical infrastructure or financing mitigation activities aiming to create alternatives to forest products and non-sustainable activities such as crop burning.

In terms of infrastructure investments, the program financed multiple types of physical benefit. For example, building schools, water distribution systems, boats for Ecotourism, and trucks for forest monitoring. These resources are provided to communities contingent on local demands and chiefdom negotiations. For livelihood income alternatives, the project funds non-forest activities aiming to reduce the need for forest-degrading activities. Activities include bee-keeping, eco-charcoal, ecotourism, and smart agriculture. For example, the

bee-keeping project distributed hives to communities to allow them to produce honey without extracting it from forests. The eco-charcoal initiative trained entrepreneurs in selective tree harvesting and in the use of more efficient carbonization kilns that reduce the amount of wood required per unit of charcoal produced. The agriculture initiative includes techniques that reduce land degradation and allow sustainable crop production. It is important to note that there is no additional financial incentive for conservation or behavior changes in regular communal areas. These initiatives aim to support and inform families in the transition to less forest-dependent income sources.

3 Data

3.1 Zambia data

3.1.1 USAID datasets

In 2015, USAID conducted baseline surveys that covered 324 villages and 4,343 households in six chiefdoms—five of which participated in the program contracts. The endline dataset, completed in June 2024, includes additional information on treatment types, such as cash transfer amounts and the specific activities delivered by the program. Both the baseline and endline surveys contain geocoded data for village and household locations, enabling spatial mapping of the surveyed communities. In addition to the surveys conducted with individuals, the organization documented geospatial datasets with conservation areas boundaries.

For this paper, I use the shapefiles of conservation areas established in partnership with chiefdoms during the program implementation. These shapefiles include regions beyond the original surveyed chiefdoms, as the implementing firm expanded the program to additional areas. In particular, new chiefdoms and conservation areas were added in after 2019. These expansions are clearly identified in the shapefile, along with the timing of additional protections. To maintain consistency in treatment, exposure grid cells located within areas included in future program expansions are excluded from all samples.

3.1.2 Chiefdom boundaries

I use the shapefile from Baldwin (2013) to define the chiefdom boundaries.[2]. This file geocodes chiefdom limits based on historical pre-colonial maps of Zambia. These boundaries ensure that the spatial delimitation of chiefdoms reflects historically accurate boundaries with cultural and political significance. Using these geocoded boundaries, I can align contemporary data with historically rooted geographic divisions of 280 chiefdoms across the country. Chiefdom boundaries are particularly valuable for considering institutional and environmental outcomes in different regions of the country, not limiting the analysis to the Eastern region. These boundaries allow for a more precise analysis of the effects by controlling for chiefdom-specific non-observable characteristics such as governance and resource management.

3.2 Satellite data

3.2.1 Tree coverage data

To measure deforestation rates, I use the Global Forest Watch dataset (Hansen et al. 2013), which provides information for 30m $\times$ 30m cells. I use two layers from this dataset, share of tree cover in 2000 and year of tree cover loss, and grouped cells into grids of approximately 1 km $\times$ 1 km ($0.01° \times 0.01°$). Figure D.1a in the appendix illustrates the baseline tree cover share per cell in 2000 for the Eastern Region. Figure D.1b identifies cells with loss of tree cover, with red cells indicating loss of tree cover in 2023. By integrating these two spatial layers, I can derive the final tree cover measurements for my analysis.

The first layer is the baseline share of tree cover for 2000, which measures the share of forest canopy at the cell level. The dataset defines a tree as any vegetation taller than 5 meters in height. Following this definition, the proportion of a cell covered by a forest canopy—the upper layer or "roof" of a forest, formed by the crowns of trees—and assigns this value to the cell. A limitation of this definition is the possibility of including plantations that meet these criteria, such as timber plantations, in my forest cover measurements. For each cell, Hansen et al. (2013) reports the share of the unit covered by the forest canopy, ranging from 0 to 100.

The second layer reports tree cover loss at the cell level. According to Hansen et al. (2013), forest loss is defined as a stand-replacement disturbance or the complete removal of tree

2. More details on the methodology and sources used to create the chiefdom boundaries can be found in Baldwin (2015), which provides a deeper exploration of the pre-colonial era maps and their relevance to modern governance structures.

canopy cover at the cell level. This implies that I am unable to track gradual degradation of tree cover in a particular cell. For example, if a cell has 50% tree cover in 2000, I cannot track if this coverage decreases to 20% in a particular period and then to zero in another subsequent year. The loss of tree cover information only indicates whether the cell was fully deforested in a given year. In the data, this is a numerical variable that indicates the year in which the cell lost its tree cover. Values range from 0 to 23, where 0 indicates no loss and 1 to 23 represents the year of loss starting from 2001. Using annual information on tree cover loss, I can create yearly deforestation measures.

One limitation of these measurements is the inability to track tree cover gains after 2000. This means that if a cell experiences an increase in tree cover after 2000, this will not be reflected in my measurements. Additionally, a cell that undergoes multiple changes, such as deforestation, reforestation, and deforestation again, will not have these variations captured by my deforestation measure.

3.2.2 Tree cover measurement.

I follow the literature to define a particular Hansen cell as forested using the canopy share threshold α. For example, if α is equal to 10, only cells with more than 10% of their area covered by forest canopy will be considered to calculate the tree cover measures of the grid. Formally, consider a set of locations $i = 1, 2, ..., n$ with n being the total number of locations considered. Each location i has j grids defined by latitude and longitude coordinates $lat_j, long_j$ that are within its location boundaries b_i such that:

$$\begin{gathered} lat_{i,Min} \leqslant lat_j \leqslant lat_{i,Max} \\ \text{and} \\ lon_{i,Min} \leqslant lon_j \leqslant lon_{i,Max} \ . \end{gathered} \tag{3.1}$$

Each of these cells j has a canopy share $cs_{j,2000}$ in the baseline year 2000. Using this information, I can define canopy share cs_i in a particular location i conditional on canopy share threshold α using the following:

$$cs_{i,2000}(\alpha) = \left(\frac{\sum_{j \in b_i} \mathbb{1}\{cs_{j,2000} > \alpha\}}{\sum_{j \in b_i} \mathbb{1}\{j \in b_i\}} \right) \ . \tag{3.2}$$

Intuitively, this equation states that forest share of a geographical location can be obtained by a simple ratio. The numerator is the sum of cells with canopy share above a specific threshold

α and that overlap with location i boundaries. The denominator is the number of cells within the location boundaries. The results corresponds to the share of forested cells within a 1km $\times$ 1km grid i.

3.2.3 Deforestation measurements

Using the tree cover and year of loss, I can obtain the annual deforested cells for a specific geographical unit. I measure this in two ways: the share of deforested cells relative to the forested cells stock in 2000 and relative to the forested cells stock in the previous year for a particular geographical grid. I do not use the Inverse Hyperbolic sine (IHS) transformation used in other deforestation papers (Cisneros and Kis-Katos 2022; Cisneros et al. 2022; Abman and Lundberg 2024) to avoid scale-sensitive estimates due to the high proportion of zeros in the data, as pointed by Chen and Roth (2024). To define the yearly deforestation rate, I add a new element to incorporate the possibility of deforestation per year. Let tl_{jy} be the binary variable equal to 1 if a particular cell j loses its tree cover in year y, zero otherwise. I also use a threshold α to limit the cells used in the deforestation measurement. The relevant cells for the calculation are the ones with canopy share above a specific threshold α. For example, if I want to calculate the deforestation rate considering forest canopy share bigger than 10%, the cell considered in both the denominator and numerator of the deforestation rate calculation are above this threshold. Using this notation, I calculate the location i of newly deforested cells in year y ($_{iy}$). Formally :

$$d_{iy} = \sum_{j \in b_i} \mathbb{1}\{tl_{jy} = 1\} \cdot \mathbb{1}\{cs_{j,2000} > \alpha\} \times cs_{j,2000} \ . \tag{3.3}$$

This measures the number of Hansen cells j with canopy share bigger than α and deforested in year y within each 1 km grid i. To consider the initial tree cover area within each 1 km block, I calculate location i deforestation rate relative to year r (D^r_{iy}) in percentage, conditional on cell canopy share threshold α as follows:

$$D^r_{iy}(\alpha) = \left(\frac{\sum_{j \in b_i} \mathbb{1}\{tl_{jy} = 1\} \cdot \mathbb{1}\{cs_{j,2000} > \alpha\}}{\sum_{j \in b_i} \mathbb{1}\{cs^r_{j,2000} > \alpha\}} \right) \times 100 \ . \tag{3.4}$$

In the numerator, I count the number of cells deforested at location i. In the denominator, I count the total number of cells categorized as forested in 2000 in year r. Both the numerator and the denominator consider a threshold α for the share of cell canopy. I put these measures

in percentage terms by multiplying by 100; therefore, all the effects are expressed in percentage points. I compute the deforestation rate relative to 2000 - r = 2000 - and to the previous year - $r = t - 1$. The deforestation relative to 2000 is computed as follows :

$$D_{iy}^{2000}(\alpha) = \left(\frac{\sum_{j \in b_i} \mathbb{1}\{tl_{jy} = 1\} \cdot \mathbb{1}\{cs_{j,2000} > \alpha\}}{\sum_{j \in b_i} \mathbb{1}\{cs_{j,2000}^{2000} > \alpha\}} \right) . \tag{3.5}$$

Which measures the proportion of the cover observed in 2000 that is lost in a particular year. This measure is informative about how grids are deforested relative to the same baseline year, which is not impacted by the non identification of reforestation that may occur after 2000. However, this measure does not reflect that losing 1 unit of forest cell may differ from when the stock of trees is getting close to zero. However, The deforestation rate relative to the previous year may reflect this differentiation on deforestation intensity. This measure is computed as follows :

$$D_{iy}^{t-1}(\alpha) = \left(\frac{\sum_{j \in b_i} \mathbb{1}\{tl_{jy} = 1\} \cdot \mathbb{1}\{cs_{j,2000} > \alpha\}}{\sum_{j \in b_i} \mathbb{1}\{cs_{j,2000}^{t-1} > \alpha\}} \right) . \tag{3.6}$$

In this measure, the denominator is updated every year by subtracting the cells lost in the year before. This implies that the same number of deforested cells will reflect a higher deforestation rate compared to D_{iy}^{2000}, reflecting a higher weight for each additional unit deforested as the forest stock decreases. I present results using both measures.

Figure D.2 in the appendix displays the distribution of tree cover share for Grids 1km$\times$1km obtained by aggregating Hansen cells in different chiefdoms. The mean of these observations ranges between 18% and 28%, indicating sparse vegetation since this region is characterized by savannas.

Supplementary data To enrich the grid-level analysis, I constructed several geospatial covariates using a variety of publicly available datasets. These include: distance to settlements (km), distance to roads (km), distance to water bodies (km), distance to the electrical network (km), altitude (meters), and land productivity. I calculated distance measures through spatial operations that compute the Euclidean distance from each grid centroid to the nearest feature in the respective shapefiles. I obtain Road and water body shapefiles from the OpenStreetMap project, and data on the electrical network from a World Bank project focused on Zambia. Settlement data comes from the GRID3 ZMB – Settlement Extents v3.0 dataset, which provides high-resolution polygons and centroid coordinates for settlements across Zambia. Using grid

coordinates and elevation data from open-access global terrain datasets I compute grid Altitude. Finally, I measured land productivity using data from the FAO's GAEZ project, specifically the agro-climatic potential yield for maize—the most commonly cultivated crop in the study region. These variables are used to control for geographic and agricultural heterogeneity in deforestation outcomes.

3.3 Summary Statistics

Table 1 provides summary statistics for all Grids located within Game Management Areas (GMAs) in Zambia, splitting these observations into five groups. The first column presents statistics for observations that are outside the Luangwa Corridor and are used as controls for the analysis. The second column shows statistics for observations within Non-Participating Chiefdoms but within the Corridor. The third column presents observations within Participating Chiefdoms, located in protected (PA) or non-protected areas (NPA), which are shown separately in columns 3 and 4.

The first row shows the average grid canopy share in 2000, indicating the average percentage of tree canopy cover in each grid by different location groups. The units of observation do not differ substantially in terms of tree cover across all five groups. Within Participating chiefdoms, protected areas have around 1.3 % more tree cover compared to non-protected areas. This suggests that blocks are not substantially different in terms of the density of trees, i.e., protected areas or Participating chiefdoms are not more forested areas in comparison to Non-Participating areas.

However, significant differences in tree cover loss statistics are present between 2001 and 2014. Non-Participating Chiefdom deforested on average around 2000 m^2 per grid relative to Participating Chiefdoms included in the program. The control groups also display higher tree cover loss, around 3000 m^2 more tree cover loss per grid. Another considerable difference exists between Protected Areas and Non-Protected areas. Deforestation rates in Protected Areas are, on average, 5 times lower than in areas not included in the protection boundaries. These simple comparisons reveal that the program delimited low deforestation areas, potentially influenced by the local authorities' endogenous choice of protection boundaries. This pattern has been found in other localities where conservation policies took place. (Giudice et al. 2019; Cisneros et al. 2022).

On the geographical characteristics, non-participating chiefdoms and controls are situ-

ated slightly higher, compared to 707.82 meters in participant chiefdoms. Locations show similar potential for maize yield varying from approximately 3170 kg/ha to 2979 kg/ha across all groups.

Another significant difference particular to Protected areas (PAs) is the significant remoteness. These grids are more remote compared to non-protected areas when considering their proximity to human settlements, roads, and energy networks. On average, blocks within PAs are located 5.3 km, 9.1 km, and 70 km farther from these infrastructures, respectively. These differences in remoteness are particularly relevant to understanding deforestation dynamics, as infrastructure availability has been shown to play a crucial role in facilitating agricultural activities and contributing to tree cover loss (Gollin and Wolfersberger, n.d.).

In summary, grids are similar in tree cover density and geographical characteristics but differ in terms of deforestation patterns and distance to infrastructure. Participating chiefdom annual tree cover loss is smaller in comparison to Chiefdoms not included in the program, and Protected Areas tend to experience less deforestation and are more remote compared to Non-protected areas.

4 Estimation Strategy

4.1 Selection of protected areas and Matching samples

The selection of conservation area boundaries within participating chiefdoms was decided in collaboration with local authorities, based on a validation process that confirmed chiefdom boundaries and land tenure arrangements. However, this selection process was not random and may have introduced bias on site selection as local leaders may strategically select low-cost areas. Because program revenues are linked to avoided carbon emissions within these areas, local authorities had a potential incentive to prioritize areas with low baseline deforestation pressure to maximize conservation gains while minimizing the actual effort required. This potential selection can limit the aggregate conservation effects due targeting of low threat regions, not allowing the program to change dynamics on tree cover loss in areas that are facing critical degradation before the program. Empirically, this form of strategic selection may threaten the causal interpretation of the program impacts as trends may not be parallel due to pre-existing land-use patterns.

To investigate the selection bias on site selection, I follow the approach of Cisneros et

al. (2022) and estimate a Logit model to examine how grid-characteristics predict the likelihood that a grid located in a participating chiefdom being allocated to conservation. The estimated equation is as follows:

$$Conservation_{ic} = \alpha + X'_{ic}\beta + \delta_c + \epsilon_{ic} \ . \tag{4.1}$$

In this model, $Conservation_{ic}$ is a dummy variable equal to 1 if the grid i is within conservation areas boundaries in a participating chiefdoms c. X_{ic} is a vector of characteristics of the grids that may correlate with the likelihood of protection status. This includes the average deforestation rate before the policy started (2001-2014), proportion of forested cells in 2000, altitude, distance to human settlements, distance to water bodies, distance to roads, distance to electricity network, distance to National Parks and a measure of potential corn yields. I add chiefdom fixed effects δ_c to control for factors related to time invariant unobservable characteristics such as cultural, and political institutions that may have impacted the protection allocation process. I estimate this model using the sample of grids within participating chiefdoms.

4.2 Matching sample

After estimating the likelihood of a grid conservation status from equation 4.1, I use estimated coefficients to compute a propensity score for each grid in the sample. The score reflects the probability that a grid is allocated to a conservation area, conditional on its observable geospatial characteristics. Using these scores, I implement a Nearest Neighbors matching algorithm without replacement and a calliper of 0.0001 to identify one matched control grid for each grid in chiefdoms impacted by the program and in nearby areas located in non-participating chiefdoms. The matched cells are found among observations outside the corridor to avoid spillover concerns. The intuition behind this procedure is to find grids that are similar in the likelihood of being allocated to conservation areas if these chiefdoms received the program.

Figures 2 and 3 display the resulting matched samples across the four groups I use to estimate the direct and spillover effects of the program in the Luangwa corridor: conservation areas, communal land, close areas ($<$ 60 km from conservation boundaries), and far areas ($\geqslant$ 60 km from conservation boundaries). It is important to note that the analysis leverages the parallel trends assumption as I use a Difference-in-Differences approach. Although this approach does not require strict exogeneity and allows for selection into treatment, it relies

on the idea that similar grids are likely to exhibit similar pre-treatment trends. By controlling for observable geospatial characteristics in the propensity score model, I aim to capture key dynamics related to deforestation and improve the validity of comparisons between treated and control units.

4.3 Participating Chiefdom outcomes

The program contract with participating chiefdoms incentivises conservation by rewarding communities for the preservation of trees within conservation areas. After consultations with local authorities, boundaries are delineated and the tree stock is used to estimate carbon offsets. Revenue from sold carbon offsets is reinvested in the Chiefdom through infrastructure and mitigation activities to reduce forest-service dependence. These investments aim to create a financial incentive that can move communities from forest degradation activities by providing alternative livelihoods. However, economic incentive theory suggests that communal land not allocated to the program, where preservation is not rewarded, may experience increases in loss of tree cover due to the leakage of deforestation from conservation areas. If this happens, this negative spillover could offset the conservation efforts implemented within the conservation areas. To test the overall effect of the program on the overall Chiefdom conservation, I first examine the effects in grids within participating chiefdoms, without considering their conservation status after the program. Next, I split these chiefdoms areas into conservation areas and communal land grids to separate possible conservation and leakage effects after program implementation. The comparison group for this analysis consists of grids in game management areas located outside the Luangwa Valley. Figure 2 illustrates the geographical distribution of these groups by conservation and communal land status for both unmatched and matched samples.

The choice of a control group outside these regions aims to reduce possible spillover concerns related to non-participating chiefdoms close to the program implementation. Due to conservation efforts from participating chiefdoms, as the 33 patrol monitoring groups documented by BioCarbon Partners (2019), other areas can be impacted by the program by implementing preservation efforts in the area. Another hypothesis is the increase in incentives for deforestation in neighboring locations, as these communities can take advantage of a possible reduction in supply of forest-related products as charcoal. Another important restriction in my sample is the location of a grid within a Game Management Area, this decision is based on the

fact that land usage laws are different across geographical layers types. Zambia is divided into National Parks, Game Management Areas, and Forest Reserves, which differ substantially in the restriction on tree cover exploration. National Parks are federally owned and have zero tolerance laws for deforestation and restrict human settlement presence, whereas GMAs and forest reserves allow it. For this analysis, I use GMAs only to avoid concerns about differences in natural resource governance.

To estimate the program's effect on Chiefdom's annual deforestation rate, I use a Difference-in-Differences (DiD) approach leveraging grid-level information (Abman and Lundberg 2024; Cisneros et al. 2022; Rico-Straffon et al. 2023; Giudice et al. 2019). The identifying assumption is that, in the absence of the program, the deforestation rates in participating chiefdoms grids would follow the same time trend as in the control group. This assumption allows for pre-existing differences in deforestation levels across groups, only requiring the trends to be parallel. Formally, contingent on cell and year fixed effects, the counterfactual deforestation trend in treated cells must be identical to the observed trend in control cells. I run the following model:

$$D^{r}_{ict} = \alpha + \beta \text{After}_t \cdot \text{Participating}_c + \eta_t + \delta_c + \epsilon_{ict} \tag{4.2}$$

In this equation, D^{r}_{ict} corresponds to the annual deforestation rate measured relative to r, 2000 or $t-1$, for Grid i in Chiefdom c in year t. $After_t$ is a dummy variable equal to 1 if year t is after 2014, when the program started. Participating_c indicates if Chiefdom c is included in the CFP contract. β_1 captures the average treatment effect of the program for grids within CFP Chiefdoms. I include time (η_t) and chiefdom (δ_c) fixed effects to control for time-invariant unobservable characteristics related to chiefdoms and particular year that may impact tree cover loss. ϵ_{ict} is an idiosyncratic error term, and errors are clustered at the chiefdom level to account for serial correlation between observations.

To estimate the yearly effects of the program and to test the parallel trends assumption for the participating chiefdom sample, I estimate an event study using the following:

$$D^{r}_{ict} = \alpha + \sum_{k} \beta_k \mathbb{1}(t = k) \text{Participating}_c + \eta_t + \delta_c + \epsilon_{ict} \ . \tag{4.3}$$

In this equation, D^{r}_{ict} corresponds to the annual deforestation rate measured relative to r for Grid i in Chiefdom c at year t. I regress these rates on yearly coefficients (β_k) before and after

the program implementation. The coefficients are yearly effects of the program on annual deforestation outcomes. To control for time-invariant unobservable characteristics that impact deforestation rates, I use time (δ_t) and chiefdom (γ_c) fixed effects. I add an idiosyncratic error term ϵ_{ict} and clustered errors at the chiefdom level to account for serial correlation.

To investigate whether the differential incentives given to communities concerning preservation in conservation areas and communal land, I estimate the differential impact of the program for these two groups. I create a dummy variable (Conservation_i) which is equal to 1 if an observation lies within the conservation areas, and a dummy (Communal_i) for grids in regular communal land areas. The identification assumption relies on the parallel trends between these two groups and controls outside the corridors. The Difference-in-Differences model is the following:

$$\begin{aligned} D^r_{ict} = \alpha + \beta_1 \text{After}_t \cdot \text{Conservation}_i + \beta_2 \text{After}_t \cdot \text{Communal}_i + \lambda \text{Conservation}_i + \\ \eta_t + \delta_c + \epsilon_{ict} \ . \end{aligned} \tag{4.4}$$

This equation is similar to 4.2, but splits Participating_c into Conservation_i and Communal_i. β_1 captures the conservation effect of the program in the delineated areas, whereas β_2 captures the possible leakage effect of the program in communal land. The coefficient λ adjusts for differences between the two groups. As previously stated, I include fixed effects, and cluster errors at the Chiefdom level.

To test for parallel trends and estimate yearly effects for these two groups, I estimate the following:

$$\begin{aligned} D^r_{ict} = \alpha + \sum_k \beta_k \mathbb{1}(t = k) \text{Conservation}_i + \sum_k \gamma_k \mathbb{1}(t = k) \text{Communal}_i + \lambda \text{Conservation}_i + \\ \eta_t + \delta_c + \epsilon_{ict} \ . \end{aligned} \tag{4.5}$$

In this event study, β_k captures the yearly effects of the program in conservation areas, and γ_k captures the yearly effects of the program in communal land. The statistical insignificance of the coefficients before the program implementation indicates parallel trends and that the coefficients from 4.4 can be interpreted as causal.

4.4 Spillover Effects on Non-Participating Chiefdoms

The intervention aimed to incentivize conservation within Chiefdoms that signed contracts with USAID. However, the delineated conservation areas are adjacent to other chiefdoms in the Luangwa Valley corridor that did not participate in the program. These neighboring Chiefdoms may experience spillover effects, either through enforcement activities such as patrol monitoring, or leakages due to shifts in deforestation activities. For instance, forest-dependent households in a non-participating chiefdom may respond to changes in the supply of forest-related products such as charcoal or timber, potentially leading to increased forest exploitation. To investigate whether the program induced such externalities, I estimate the program effects on deforestation rates in non-participating chiefdoms located within the corridor.

Following a similar approach to the analysis of the participating chiefdoms (4.2), I first estimate the average effect of the program for grids within non-participating chiefdoms using a Difference-in-Differences specification:

$$D^{r}ict = \alpha + \beta \text{After}_t \cdot \text{Non-Participating}_c + \eta_t + \delta_c + \epsilon_{ict} \ . \tag{4.6}$$

In this equation, D^{r}_{ict} represents the annual deforestation rate for grid i in Chiefdom c at year t, relative to r. The variable $\text{Non-participating}_c$ indicates whether a Chiefdom is located within the corridor but was not included in the program contract. The interaction term captures the potential spillover effect of the program on these Chiefdoms (β). Similar to other regressions, I include year (η_t) and Chiefdom (δ_c) fixed effects, and cluster standard errors at the Chiefdom level.

To explore the yearly effects and validate the parallel trends assumption for this group, I estimate the following event study specification:

$$D^{r}ict = \alpha + \sum_{k} \beta_k \mathbb{1}(t = k)\text{Non-Participating}_c + \eta_t + \delta_c + \epsilon_{ict} \ . \tag{4.7}$$

This equation estimates yearly effects of the program in Non-participating Chiefdoms relative to the baseline period, 2014. Insignificant coefficients in the pre-program period would support the parallel trends assumption, allowing for causal interpretation of post-treatment coefficients.

To explore heterogeneity in spillover effects within Non-Participating Chiefdoms, I clas-

sify grids based on their proximity to the nearest Protected Area. I define two groups: Close areas, located $<$ than 60 km from the boundary of conservation areas, and Far areas, located beyond this threshold. This allows me to examine whether spillovers are more pronounced in areas closer to the intervention. Figure 3 illustrates the geographical distribution of these groups. The DiD specification is:

$$D^{r}ict = \alpha + \beta 1After_t \cdot Close_i + \beta_2 After_t \cdot Far_i + \lambda Close_i + \eta_t + \delta_c + \epsilon_{ict} \ . \tag{4.8}$$

In this regression, β_1 captures the spillover effect in Close areas, and β_2 captures the effect in Far areas. The coefficient λ adjusts for differences between the two geographical groups. As in previous models, I include fixed effects and cluster errors at the Chiefdom level.

Finally, I estimate an event study to capture the yearly evolution of spillover effects by distance to the program boundaries:

$$D^{r}ict = \alpha + \sum_k \beta_k \mathbb{1}(t = k) Close_i + \sum_k \lambda_k \mathbb{1}(t = k) Far_i + \lambda Close_i + \eta_t + \delta_c + \epsilon_{ict} \ . \tag{4.9}$$

Here, β_k measures yearly effects for Close grids, while λ_k measures effects for Far grids. Pre-treatment insignificance would support the parallel trends assumption, while post-treatment differences reveal the extent and timing of spatial spillovers.

5 Results

In this section, I present the results on the selection of conservation areas and the impact of the program on annual tree cover loss. First, I show the results related to the selection of conservation areas. Second, I discuss the effects on deforestation trends for participating chiefdoms, conservation areas, and communal land within these chiefdoms. Third, I show the spillover effects in neighboring chiefdoms that did not participate in the program. Finally, I assess how the program impacts vary based on grid-level characteristics, shedding light on potential sources of heterogeneity in treatment effects.

5.1 Selection of conservation areas

Table 2 displays the results of equation 4.1. The first column shows results from the logit estimation using measures of 1km x 1km grid average annual deforestation rate in % (2001-2014),

proportion of forested cells, and measures of distance to electrical infrastructure, National Parks, roads, human settlements, and water bodies. The second column adds grid altitude and productivity measure by corn potential yields. The third column includes chiefdom fixed effects. All specifications show that higher deforestation rates before policy are strongly associated with a lower probability of protection, suggesting a strategic choice of low-deforestation places. Another important pattern is the significance of characteristics related to the remoteness of the grid. The distance to roads, electrical networks, water bodies, and human settlements is positively associated with protection, suggesting that more remote areas were more likely to be selected. This aligns with the hypothesis that protection was targeted in areas facing lower pressure of deforestation and harder access. These patterns remain consistent in column (3) after adding chiefdom fixed effects. Overall, the results suggest that communities selected areas to be protected where there were lower deforestation patterns and lower opportunity costs of conservation.

To illustrate how different potential controls and grids within the corridors might be, I plot the Propensity score distribution across these groups, before and after the matching procedure. Figure 4 shows these distributions for conservation areas (CA) and communal land (CL). Two key observations about these distributions. First, Potential controls present a distribution highly concentrated to the left, indicating the unlikelihood of treatment as predicted by the logit model. This indicates that the unmatched sample as a control group for conservation areas, the comparison includes grids with low conservation probability. Second, the nearest-neighbor matching procedure that finds one similar pair for each grid in participating chiefdoms overlaps the propensity score distributions for both groups, as seen in Figures 4b and 4d. The geographical distribution of Figure 2 maps the unmatched and matched samples. Similarly, figure D.3 plots the distributions for the close and far grids within non-participating chiefdoms. Figure 3 maps the sample for the non-participating chiefdoms.

Table 1 shows that grid cells differ systematically across groups in geospatial characteristics such as altitude and distance to infrastructure. Panel A presents the full unmatched sample: Conservation Areas and Communal Land are located at lower altitude and farther from roads and settlements than outside-corridor controls, consistent with program placement in remote, forested areas. Panel B shows that propensity score matching reduces imbalance on several covariates, but substantial differences in altitude and distance to the electrical network persist after matching. This reflects a fundamental geographical separation between the Lu-

angwa Valley, where participating chiefdoms are concentrated, and the surrounding plateau that provides most of the control pool, a pattern documented in other conservation papers (Giudice et al. 2019; Cisneros et al. 2022).

It is important to note that my identification strategy assumption does not require treated and control grids to be similar in observable characteristics, but that they would have followed parallel trends in deforestation rates in the absence of the program. To address selection concerns and assess sensitivity to control group composition, I present results for both unmatched and matched samples throughout.

5.2 Participating chiefdoms

The estimates of the effect of the program on participating chiefdom annual deforestation rate are in Table 3. I estimate regressions 4.2 and 4.4 using two different outcome measures, the annual deforestation rate relative to 2000 (D^{2000}), and relative to the previous year (D^{t-1}). The table is split into two parts to highlight the differences in the sample. Columns 1 to 4 present results using the raw sample of potential controls outside the corridors (Figure 2a), and Columns 5 to 8 using the matched samples (Figures 2b and 2c).

The table shows a consistent pattern of tree cover loss reduction effects in conservation areas. When using the raw sample as a comparison group, I find a reduction of 0.119 percentage points in the deforestation rate relative to forest stock in 2000. This coefficient is more negative when using the annual deforestation relative to the previous year, indicating a reduction of 0.146 percentage points. The same pattern is observed using the matched sample, but the magnitudes are bigger. The results indicate that CAs have a 0.16 percentage point reduction in their annual deforestation rate relative to 2000, and 0.187 relative to the previous year.

To interpret the magnitudes of these coefficients, I compare the estimates to the average increase in the matched counterfactual group's average deforestation rate in Table 4. According to these comparisons, the program avoided 87% - 88% of the deforestation increase observed in the counterfactual. Translating these coefficients to hectares, the conservation effects correspond to yearly avoided deforestation of 291-325 ha and 2,620–2,925 ha from 2015 to 2023. To obtain these estimates, I use the number of conservation area cells, their mean of forested cells, and the avoided deforestation effects. The magnitudes are comparable in scale to deforestation increases documented in similar geospatial panel settings, as Abman and Lundberg

(2024), which finds oil palm market access contracting increased forest loss rates by 116–132% of the pre-treatment mean in Ghana, suggesting that well-enforced conservation programs can offset deforestation pressures of comparable magnitude to those induced by agricultural expansion. It is also important to note that although the magnitudes are large, the program took place in areas with a very low deforestation rate, which may explain the almost complete control of deforestation activities after the program.

However, no leakage effects on communal land and overall impact at the Chiefdom level are observed as all the coefficients are statistically insignificant. The absence of leakage can be due to two things. First, conservation areas were not used intensively by the communities, as shown in table 1, which may explain why the deforestation leakage should be small if any. The heterogeneous effects analysis, shown later, indicates that the program effects were driven by the reduction in deforestation rates close to roads and villages. As conservation areas are difficult to access and low populated regions, agents that were restricting their forest activities after the program may be constrained in their capacity to deforest outside the conservation boundaries. However, if they do start deforesting more in Non-Protected Areas, the proportion of this event may not be enough to change the region's deforestation trends after the policy. The size of CA relative to CL may not be enough to change the aggregate trends of participating chiefdoms. Another explanation is compensation through the implementation of mitigation activities within communities. Agents living in participating chiefdoms may be able to acquire income from non-deforesting services financed by the program, such as forest patrol, eco-tourism, and bee-keeping production, which may reduce their incentives for tree cover loss.

The parameters discussed before can be interpreted as causal if the parallel trends assumption holds, indicating that participating chiefdom areas and outside the corridor GMAS would follow the same trend in the absence of the program. Event study results from equation 4.5 and 4.9 can be seen in figures 5 and 6. These estimates show that both the deforestation rate relative to 2000 and previous years display a very similar pattern in terms of coefficient magnitudes and significance. The matched sample coefficients before policy (pre-trends) look closer to the zero line, indicating visual improvement in the parallel trends assumptions. The yearly coefficients after the implementation of the policy suggest that the reduction in the yearly annual deforestation rate observed in conservation areas is statistically significant after 2019, after the pilot period (2015-2018) and when PES payments started as the carbon credit market were

established. The start of yearly payments after 2019 may have brought additional benefits to communities and improved trust in contract conditions, incentivizing consistent conservation. Another possibility is the transition time necessary from forest dependent services that took place before 2019 and during the first cycle of the program.

These results are robust to different matching approaches, as seen in 6 in Panel A. The coefficients are very similar in magnitude across all the ATE effects estimates using different PSM calipers or allowing for more than one pair for each grid in the treatment groups. One important note is the insignificant coefficient for the Mahalanobis matching, which may be due to overfitting on observable characteristics as the grids are located in systematically different geographies as discussed before. I also test how changing treatment year to 2019 and omitting time chunks from the pilot (2014-2018) and after 2019 to understand whether effects are driven by PES initiation and find significant results across specifications. Table D.7 shows that the main coefficients of -0.141 pp and -0.157 pp are more negative, around 0.23 pp to -0.26 pp, when omitting the pilot period and changing the initial year of treatment. This is a consequence of the stronger yearly coefficients after 2019 discussed before.

I also show parallel trends assumptions tests in tables D.1 and D.2 available in the appendix. The first and third columns show the difference between conservation areas and communal land of the program before the program implementation, estimated using the average yearly coefficients effect of the program before 2015. Even though there is no evidence of statistically different trends before the program in the unmatched sample, the matched sample has coefficients closer to zero.

5.3 Non-participating chiefdoms

Similarly to the estimates on the participating areas, Table 5 presents these estimates for non-participating areas. I estimate regressions 4.8 and 4.9 using two different outcomes measures, the annual deforestation rate relative to 2000 (D^{2000}), and relative to the previous year (D^{t-1}). The tables follow the same division of unmatched sample from columns 1 to 4 (Figure 3), and matched samples from columns 5 to 8 (Figures 3b and 3c). I define close and far categories by grid distance from the conservation areas boundaries. Close areas are < 60 km from the CA boundaries, and far areas are $\geqslant 60$ km from the boundaries. This threshold is the mean distance of Non-Participating chiefdoms grids to Conservation Areas.

I find conservation spillover in closer areas, which faces reductions in the annual de-

forestation rate of tree cover loss. The raw sample results coefficients indicate a reduction of 0.1 percentage points on the deforestation rate relative to forest stock in 2000, which is slightly less negative when using the annual deforestation relative to the previous year, -0.130. When comparing with the matched sample, the magnitudes are bigger in absolute terms. Closer Areas have a 0.173 percentage points reduction in their annual deforestation rate relative to 2000, 0.194 relative to the previous year. This suggests that the implementation of the program in areas in the corridor may have induced conservation in neighboring areas. One hypothesis for this impact is the increase in monitoring and enforcement from participating chiefdoms, which can behave as protectors of the region due to incentives to protect their conservation boundaries. Another possible explanation is knowledge sharing between these regions, which may lead them to transition to less forest-dependent services and products. Finally, a general equilibrium effect may disincentivize non-participating chiefdoms to deforest. As the supply reduction of forest-related services may increase unit prices, communities can make the same income with less production. Far areas do not present significant effects.

I use the same approach used for conservation areas to interpret the magnitudes of these coefficients by comparing the ATE estimated with the average increase in the matched counterfactual group's deforestation rate. Close areas reduction corresponds to a reduction of 58.7% - 60.7% of the deforestation increase observed in the counterfactual. Similarly to the conservation areas' effects, I use the mean information of these cells and the estimated ATE effect to measure avoided tree cover loss in hectares. In hectares, the spillover effects correspond to yearly avoided deforestation of 462–541 ha and 4,160–4,870 ha from 2015 to 2023.

Figure 6 shows the coefficients of event studies obtained using the close and far areas samples to run equation 4.9. These figures aim to visually investigate the yearly effects of the program and the parallel trends assumptions between these areas and unmatched and matched controls outside the Luangwa corridor. Similar to the participating chiefdoms, the deforestation rate relative to 2000 and previous years displays similar patterns in terms of coefficient magnitudes and significance. The matched sample does display a better parallel trend, with coefficients concentrated around zero before the policy. The reduction pattern after 2018 is the same as in conservation areas, indicating that neighboring regions follow similar patterns, which indicate spillover effects of the increasing effort to preserve from participating areas. Robustness tests in table 6 Panel B show that the estimated ATE spillover is robust and

similar in magnitude across different propensity score matching algorithms.

5.4 Heterogeneous Effects

Figure 7 presents event study estimates from tables D.3 and D.4 to examine the heterogeneous effects of the policy. The analysis is disaggregated by conservation areas and closed areas, as these are the only groups with statistically significant coefficients. Figure 7a shows the estimates using the deforestation rate relative to the baseline tree cover year 2000 (D^{2000}) and figure 7b shows results using the rate relative to the previous year (D^{t-1}).

Conservation areas grids have a statistically insignificant coefficient associated to proximity to CA borders, and mean tree cover in 2000. This suggests that the program did not have a differential impact for more forested grids and those relatively more remote relative to the Protection borders. However, I find differential effects for distances to roads and settlements, reaffirming the role of accessibility on deforestation dynamics. The coefficient associated to settlement proximity indicates that the program was less effective in more remote places, associating one additional km farther to a coefficient of 0.022 percentage points smaller for the deforestation rate reduction. Additionally, one km farther from roads is associated with a smaller reduction in deforestation rate of 0.006 percentage points. This indicates that the effects of the program are driven by grids that are closer to villages and road access, indicating the conservation effects on places with deforestation pressure. These findings are on line with other conservations studies which find that infrastructure proximity increase deforestation and that consevation program are more effects on places where threats were higher (Giudice et al. 2019; Cisneros et al. 2022; Araujo, Assunção, and Bragança 2023).

For close areas, I find no heterogeneous impacts for the all group. However, I find heterogeneous effects for different distance groups within close areas, suggesting a decreasing distance gradient of the conservation spillover effect. Tables D.5 and D.6 show a decreasing effect of the program for grids in non-participating chiefdoms that are farther from conservation area boundaries. Grids from 0-15 km have a decrease of 0.19 pp, from 15-30 km the effect is -0.15 pp, from 30-45 km -0.145 pp, and insignificant results beyond 45 km.

6 Conclusion

This paper investigates the impacts of one of the largest REDD+ conservation programs in Africa, implemented by USAID and BioCarbon Partners in Eastern Zambia. Using this con-

servation initiative, I investigate how selection bias due to conservation financial incentives operates in regions where land governance is based on customary institutions. I also investigate to what extent such interventions can effectively reduce deforestation and its spillover effects in neighboring areas.

To perform this analysis, I constructed a geospatial panel dataset with a high spatial resolution (1km $\times$ 1km) spanning from 2001 to 2023. I combine remote sensing data on tree cover loss with administrative information on newly established conservation areas, historical chiefdom boundaries, and various geospatial variables. The empirical strategy employs a Difference-in-Differences framework that exploits variation across time and space using matched samples to reduce concerns with selection bias. I further disaggregated participating chiefdom areas by conservation areas and communal land, and explored potential spillovers into neighboring non-participating regions close and far from the conservation boundaries. Heterogeneous effects are examined based on proximity to infrastructure, settlement density, and land characteristics.

The analysis reveals three patterns. First, conservation area boundaries were systematically placed in remote, low-deforestation locations, consistent with local authorities minimizing conservation effort to maximize program revenues. Despite the selection, I found an estimated 87–88% reduction in annual deforestation within Conservation Areas. Moreover, I find no evidence of deforestation leakage into adjacent communal land within participating chiefdoms, suggesting that the program did not displace forest-degrading activities within contracted boundaries. Second, non-participating chiefdoms located close to conservation boundaries experienced reductions of around 58–60% in annual deforestation rates. Third, these spillovers and direct effects are strongest in grids closer to roads and human settlements, where deforestation pressure is typically highest, indicating that the program was most effective precisely where conservation effort matters most.

To provide additional evidence of the robustness of my results, I show that these results survive using other matching algorithms, changing the treatment year and omitting important time chunks of the treatment period. Moreover, I show event study estimates and parallel trends tests, which indicate that groups were following parallel trends before the policy implementation, and these effects are causal estimates. Robustness and sensitivity analysis following Roth (2022) and Rambachan and Roth (2023) shows that the estimated effects are robust to small violations of the parallel trends assumption, and that any confounding pre-existing

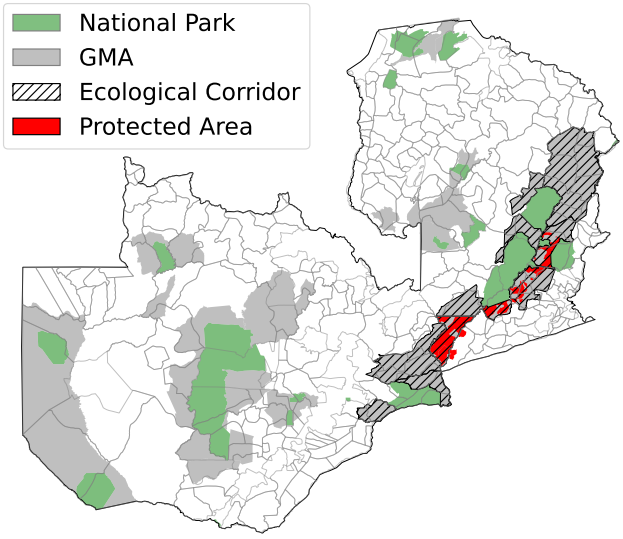
trend would run in the opposite direction of the estimated effects.

These findings have important policy implications. The spillover results show that the influence of conservation programs can extend well beyond formally contracted boundaries, suggesting that standard evaluations focused exclusively on designated conservation areas may underestimate program impacts. The evidence of strategic site selection points to a need for improved incentive structures when establishing conservation boundaries. Because program revenues are tied to avoided deforestation within designated areas, local communities have an incentive to prioritize low-threat zones that minimize conservation effort. Targeting mechanisms that condition eligibility or payment rates on baseline deforestation pressure can redirect conservation effort toward high-risk areas and may improve aggregate effects.

This study contributes to the broader literature on environmental policy, land governance, and the design of conservation incentives. It underscores the importance of integrating spatial heterogeneity, local governance dynamics, and behavioural responses into the design and evaluation of conservation programs. Future conservation initiatives should carefully consider where interventions take place and how site selection decisions interact with the underlying land-use pressures and incentive mechanisms. The spillover results suggest that conservation tied to proper incentive design and enforcement can foster better forest management well beyond the boundaries of formally designated areas.

Figures and Tables

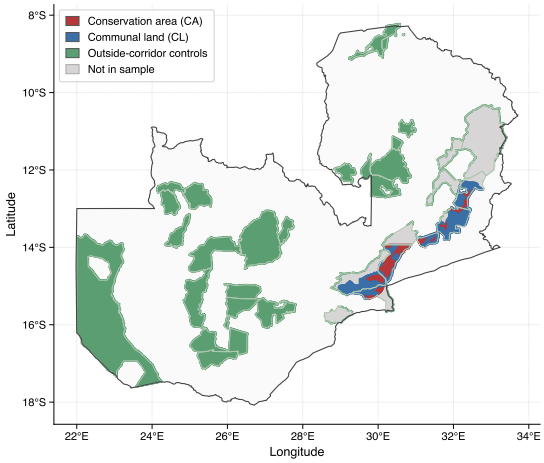
Fig. 1. Game Management Areas, National Parks and REDD+ Protected areas in Zambia



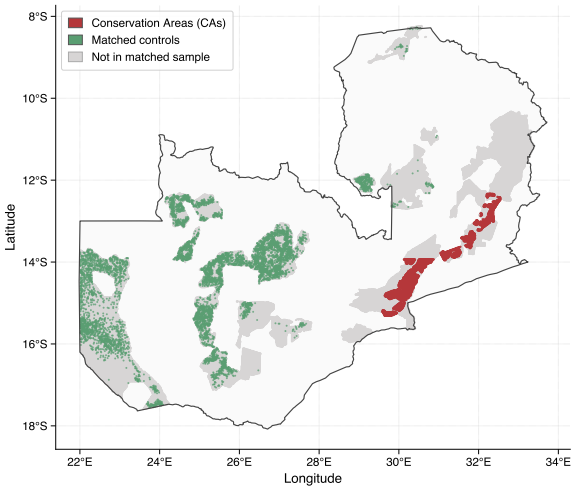
(a) Eastern Province

Note: This map shows Zambia and its chiefdom boundaries with additional layers of Game Management Areas (GMAs), National Parks, and the program conservation areas in Zambia. The green layer represents the Natural Parks, the gray layer shows the GMAs and the red layer indicates the conservation areas defined by the CFP program. The hatched area indicates the Luangwa Valley, an important biodiversity location in Eastern Zambia, targeted by the program.

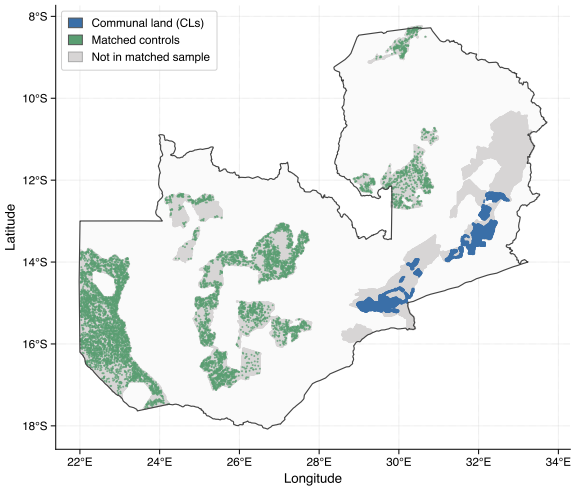
Fig. 2. Participating chiefdom samples



(a) Unmatched sample



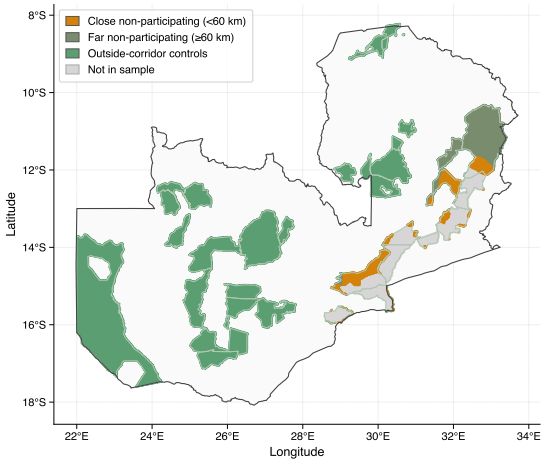
(b) Conservation areas matched sample



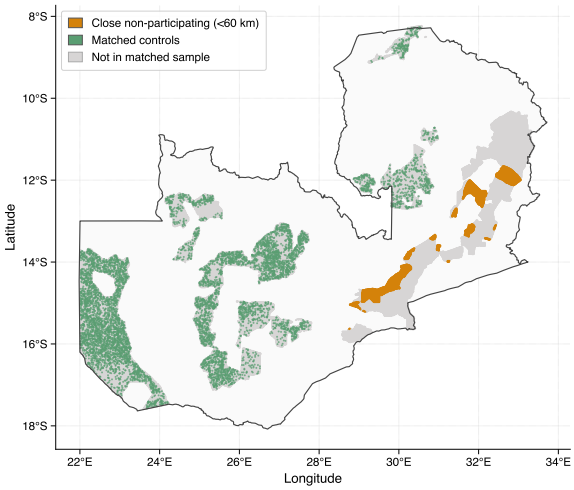
(c) Communal land matched sample

Note: These maps show the spatial distribution of cells included in the analysis for Participating Chiefdoms. Panel (2a) shows the full unmatched sample, with Conservation Area (CA) cells and Communal Land (CL) cells are distinguished by color, alongside the outside corridor control cells. Panels (2b) and (2c) show the retained cells after propensity-score matching (nearest-neighbor, 1:1 without replacement, caliper = 0.0001) for the CA and CL comparisons, respectively.

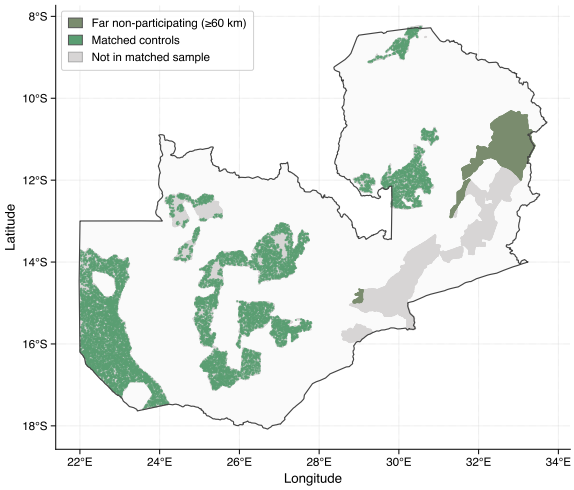
Fig. 3. Non-Participating chiefdom samples



(a) Unmatched sample



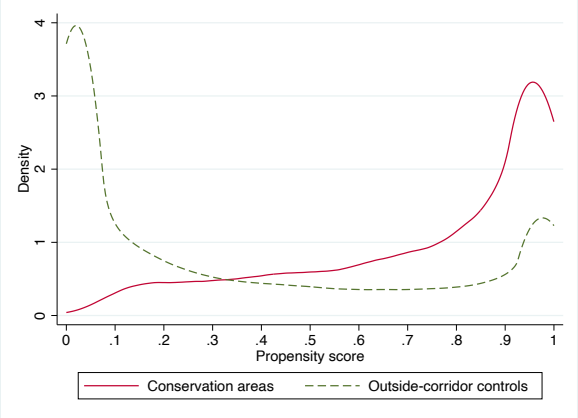
(b) Conservation areas matched sample



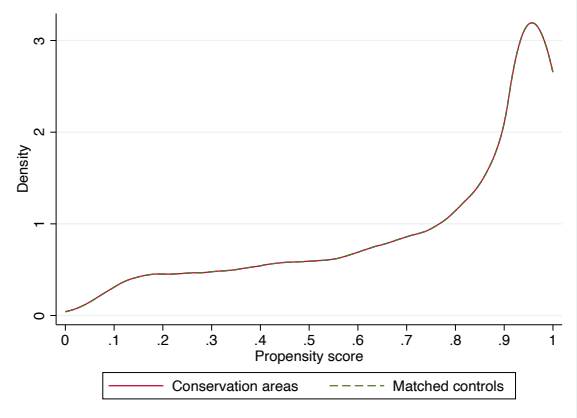
(c) Communal land matched sample

*Note:*These maps show the spatial distribution of cells included in the analysis for Non-Participating Chiefdoms. Panel (3a) shows the full unmatched sample, with close cells ($\leqslant$ 60 km from Conservation Area boundaries) and far cells ($\geqslant$ 60 km) distinguished by color, alongside outside-corridor control cells. Panels (3b) and (3c) show the retained cells after propensity-score matching (nearest-neighbor, 1:1 without replacement, caliper = 0.0001) for the close and far comparisons, respectively.

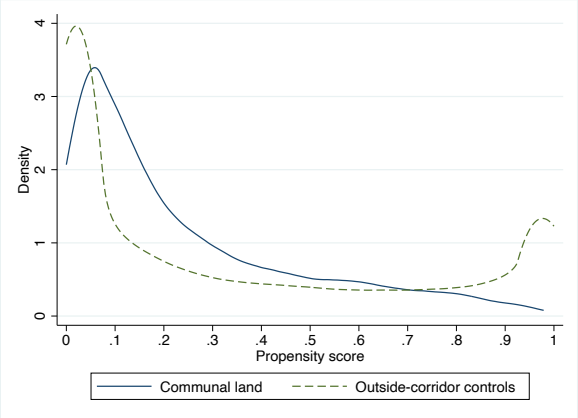
Fig. 4. Grid propensity score distributions for participating chiefdoms



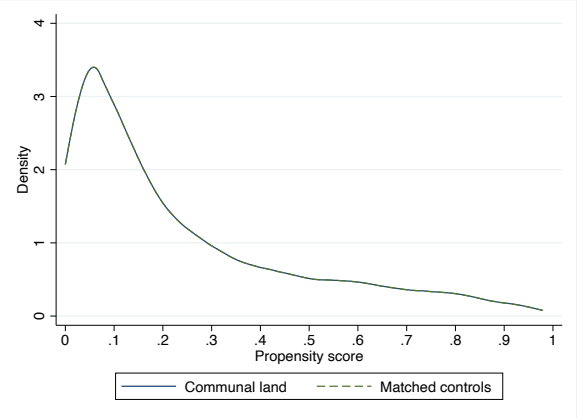
(a) CA: Before



(b) CA: After



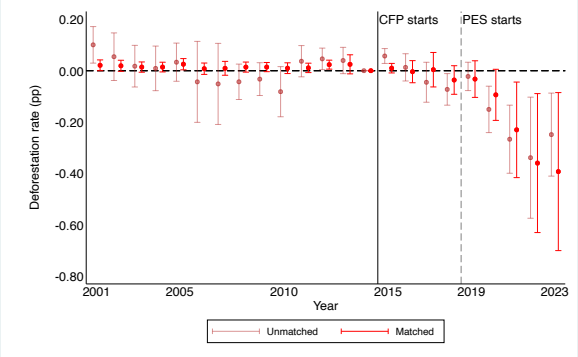
(c) CL: Before



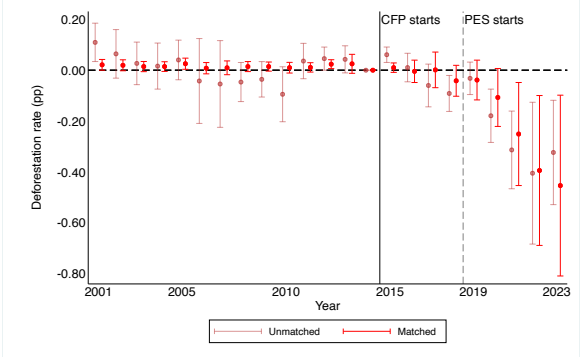
(d) CL: After

Note: Panels 4a and 4b show propensity score distributions for Conservation Area (CA) cells and their matched controls before and after matching. Panels 4c and 4d show the same for Communal Land (CL) cells. Propensity scores are estimated from equation 4.1 using pre-policy deforestation rates, altitude, distance to roads, distance to human settlements, distance to the electrical network, and baseline forest cover.

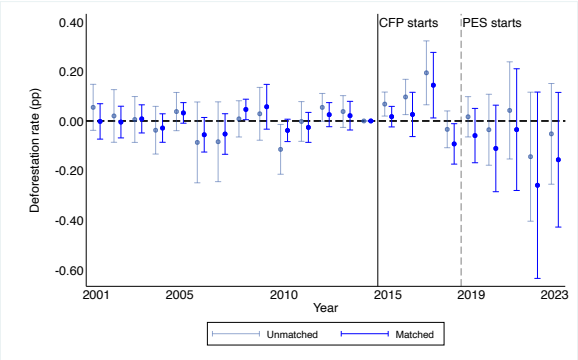
Fig. 5. Event Study Coefficients: Conservation Areas and Communal Land



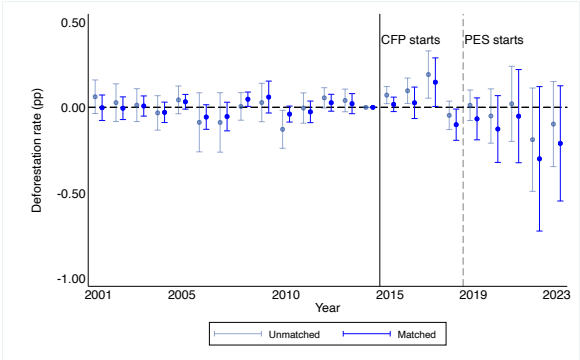
(a) CA: D^{2000}



(b) CA: D^{t-1}



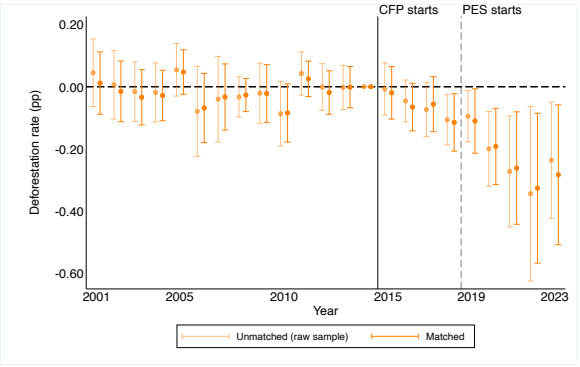
(c) CL: D^{2000}



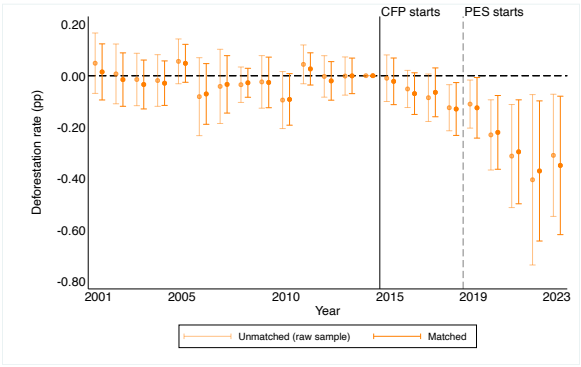
(d) CL: D^{t-1}

Note: This figure plots event-study coefficients from equation 4.5, estimated on the PSM-matched sample. The x-axis shows years relative to program start (2015 = year 0); the reference year is 2014. Panels 5a and 5b show yearly treatment effects for Conservation Areas (CA) using D^{2000} and D^{t-1}, respectively. Panels 5c and 5d show the same for Communal Land (CL) within contracted chiefdoms. Coefficients near zero in the pre-treatment period (2001–2014) support the parallel trends assumption for the matched sample. The acceleration of negative effects after 2018 is consistent with the introduction of PES carbon market payments in 2019. Shaded bands correspond to 95% confidence intervals with standard errors clustered at the chiefdom level.

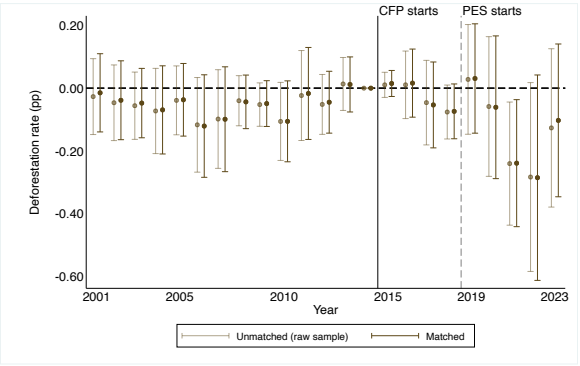
Fig. 6. Event Study Coefficients: Close Areas and Far areas



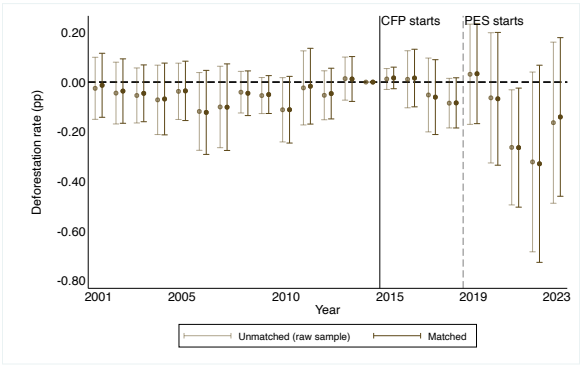
(a) Close: D^{2000}



(b) Close: D^{t-1}



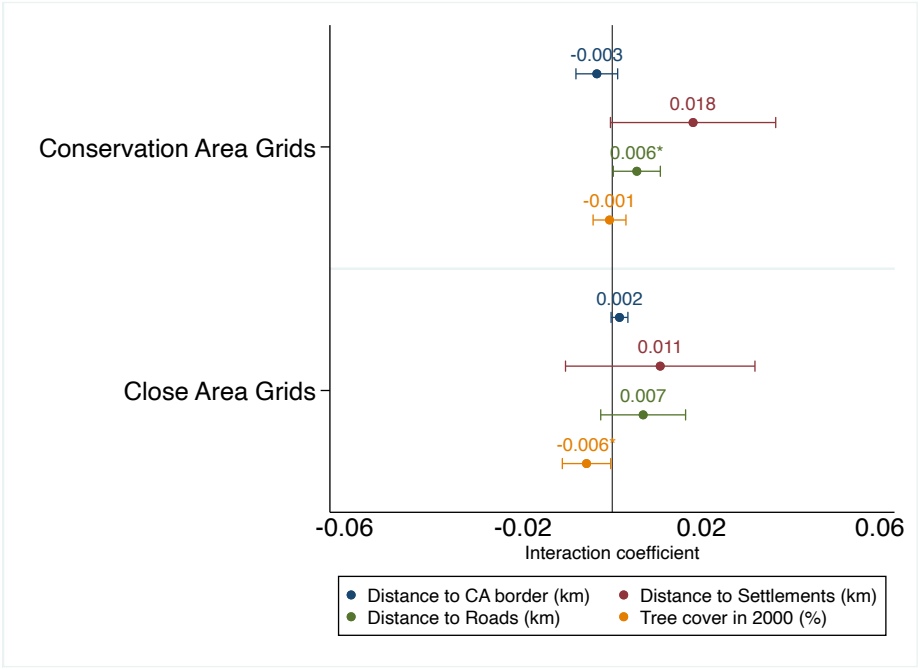
(c) Far: D^{2000}



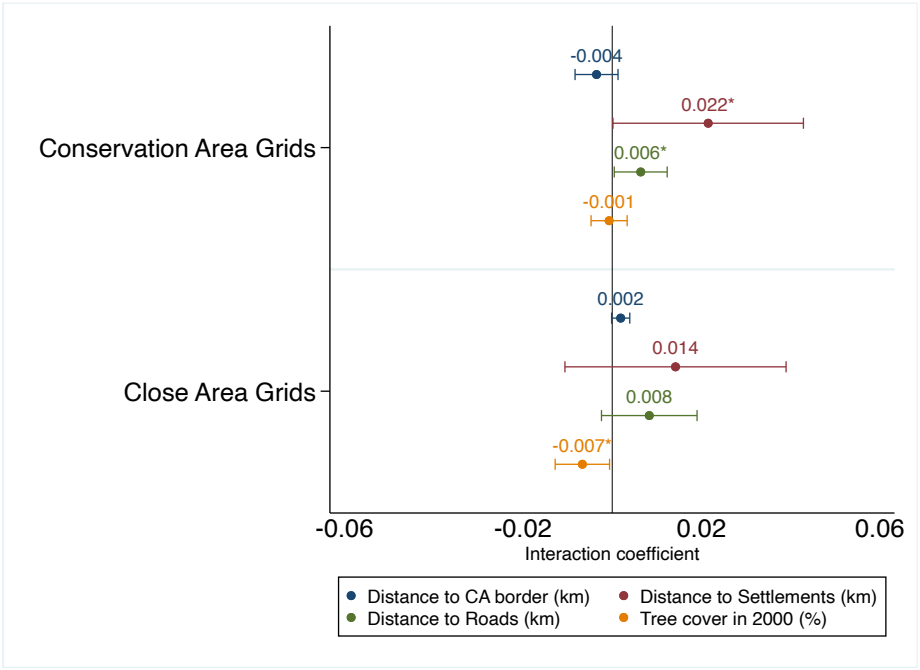
(d) Far: D^{t-1}

Note: This figure plots event-study coefficients from equation 4.9, estimated on the PSM-matched sample for Non-Participating Chiefdoms. The x-axis shows years relative to program start (2015 = year 0); the reference year is 2014. Panels 6a and 6b show yearly treatment effects for close areas ($\leqslant$ 60 km from Conservation Area boundaries) using D^{2000} and D^{t-1}, respectively. Panels 6c and 6d show the same for far areas ($\geqslant$ 60 km). The negative pre-period coefficients visible in the far area panels (2012–2014) reflect mild pre-existing differences that persist despite matching; far-area estimates should therefore be interpreted with caution. Shaded bands correspond to 95% confidence intervals with standard errors clustered at the chiefdom level.

Fig. 7. Heterogeneous effects estimates for Conservation areas and Close areas



(a) D^{2000}



(b) D^{t-1}

Note: This figure presents heterogeneous treatment effects of the CFP estimated on the PSM-matched sample. Panels 7a and 7b show results for grids within Conservation Areas and Close areas using D^{2000} and D^{t-1} as outcomes, respectively. In each panel, four interactions are plotted: the treatment indicator interacted with (i) distance to the Conservation Area boundary (in km), (ii) distance to human settlements (in km), (iii) distance to roads (in km), and (iv) the grid's baseline share of tree canopy cover in 2000. Each coefficient represents the estimated change in the annual deforestation rate associated with a one-unit increase in the respective interaction covariate. Standard errors are clustered at the chiefdom level. Error bars correspond to 95% confidence intervals.

Table 1: Corridor GMAs summary statistics by group

			Participating		
	Outside Controls	Non-Participating	CA and CL	CA	CL
Panel A: Before PSM matching					
Cell canopy share in 2000 (%)	22.416	24.994	22.889	23.632	22.205
	(16.725)	(9.217)	(6.876)	(7.316)	(6.369)
Tree cover loss in m^2 (2001–2014)	17514.659	12430.953	8021.574	1883.995	13678.033
	(50699.733)	(34543.932)	(25569.283)	(7930.537)	(33641.040)
Average annual tree cover loss in m^2	1251.047	887.925	572.970	134.571	977.002
	(3621.409)	(2467.424)	(1826.377)	(566.467)	(2402.931)
Average annual rate of tree loss in % (2001–2014)	0.125	0.064	0.041	0.010	0.070
	(0.345)	(0.179)	(0.131)	(0.041)	(0.172)
Altitude (m)	1108.717	838.737	709.144	696.967	720.367
	(86.224)	(243.591)	(173.599)	(163.660)	(181.567)
Maize Potential Yield (kg/acre)	3173.675	2968.944	3090.504	3100.424	3081.361
	(173.565)	(125.935)	(94.689)	(80.369)	(105.372)
Distance to human settlements (km)	3.818	3.769	3.859	5.979	1.906
	(4.662)	(3.241)	(3.391)	(3.395)	(1.877)
Distance to roads (km)	6.835	4.583	7.101	9.579	4.818
	(6.866)	(4.256)	(5.573)	(5.699)	(4.345)
Distance to Electrical network (km)	69.486	38.205	66.492	78.471	55.452
	(40.036)	(23.431)	(38.890)	(37.917)	(36.433)
Observations	98656	22563	15125	7254	7871

Panel B: After PSM matching (caliper = 0.0001, 1:1 NR)

	CA comparison		CL comparison	
	Matched Controls	CA	Matched Controls	CL
Cell canopy share in 2000 (%)	30.743	23.639	20.436	22.207
	(17.329)	(7.317)	(14.905)	(6.369)
Tree cover loss in m^2 (2001–2014)	2325.768	1889.227	7847.093	13694.975
	(7636.888)	(7947.251)	(21206.925)	(33659.064)
Average annual tree cover loss in m^2	166.126	134.945	560.507	978.213
	(545.492)	(567.661)	(1514.780)	(2404.219)
Average annual rate of tree loss in % (2001–2014)	0.013	0.010	0.052	0.070
	(0.045)	(0.042)	(0.131)	(0.172)
Altitude (m)	1143.365	697.002	1095.093	720.354
	(78.325)	(163.713)	(85.786)	(181.553)
Maize Potential Yield (kg/acre)	3205.965	3100.375	3200.647	3081.326
	(146.226)	(80.421)	(175.398)	(105.363)
Distance to human settlements (km)	6.523	5.985	2.541	1.904
	(4.015)	(3.399)	(2.359)	(1.876)
Distance to roads (km)	9.583	9.579	6.214	4.815
	(7.211)	(5.703)	(6.192)	(4.344)
Distance to Electrical network (km)	92.506	78.471	65.536	55.433
	(35.001)	(37.923)	(37.438)	(36.418)
	7222	7222	7861	7861

Note: This table presents summary statistics for cells in the study area. Observations are divided into Outside Controls (cells outside the Luangwa valley corridor), Non-Participating chiefdoms, and Participating chiefdoms. Participating chiefdoms are split into Conservation Areas (CA) — land set aside for conservation under the contract — and Communal Land (CL) — remaining chiefdom area outside the CA boundary. Panel A reports statistics for the full, unmatched sample. Panel B reports statistics after my preferred propensity-score matching algorithm (caliper = 0.0001, 1:1 nearest-neighbor without replacement), separately for the CA and CL comparisons. The table presents the mean value for 1 km×1km grid cells within each group. Standard deviations are presented in parentheses.

Table 2: Estimates of Logit regression Marginal effects on grid protection status

	(1)	(2)	(3)
Average annual Deforestation rate in % (2001-2014)	-0.73***	-0.71***	-0.63***
	(0.07)	(0.07)	(0.06)
Tree cover share in % (2000)	0.00***	0.01***	-0.00***
	(0.00)	(0.00)	(0.00)
Distance to Electrical networks (km)	0.00***	0.00***	0.01***
	(0.00)	(0.00)	(0.00)
Distance to National Parks (km)	0.00***	0.00*	0.00*
	(0.00)	(0.00)	(0.00)
Distance to Roads (km)	0.01***	0.01***	0.01***
	(0.00)	(0.00)	(0.00)
Distance to Settlements (km)	0.07***	0.07***	0.07***
	(0.00)	(0.00)	(0.00)
Distance to water bodies (km)	-0.01***	-0.01***	-0.01***
	(0.00)	(0.00)	(0.00)
Altitude in m		-0.00***	0.00
		(0.00)	(0.00)
Maize Potential Yield (kg/acre)		0.00***	0.00
		(0.00)	(0.00)
Observations	15,125	15,125	15,047
Chiefdom FE	No	No	Yes

*Note: This table presents marginal effects from a logit regression where the dependent variable equals one if a 1 km $\times$ 1 km grid cell is designated as a Conservation Area (CA). The sample is restricted to cells within Participating Chiefdoms. Column (1) includes only pre-policy deforestation rates as covariates. Column (2) adds geographic controls (altitude, distance to roads, settlements, and electrical network). Column (3) further adds baseline forest cover (share of forested area in 2000). *,**,*** indicates 10%, 5% and 1% statistical significance levels respectively.*

Table 3: Estimates of CFP's impact on Participating Chiefdom's Annual Deforestation rate

	Raw Sample				*Matched Sample*					
	D^{2000}		D^{t-1}		D^{2000}			D^{t-1}		
Contracted Chiefdom	-0.049		-0.073		-0.096			-0.112		
	(0.065)		(0.074)		(0.064)			(0.073)		
Conservation Area		-0.126**		-0.156**		-0.141**			-0.157**	
		(0.058)		(0.067)		(0.055)			(0.062)	
Communal Land		0.022		0.006			-0.057			-0.073
		(0.066)		(0.076)			(0.077)			(0.088)
R^2	0.04	0.04	0.04	0.04	0.04	0.04	0.05	0.04	0.04	0.05
Observations	2,358,788	2,358,788	2,358,788	2,358,788	650,716	320,689	343,022	650,716	320,689	343,022
Chiefdom FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Matched Sample	No	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
# Chiefdoms	85	85	85	85	82	60	80	82	60	80
WCB p-value	0.476	0.068 / 0.737	0.358	0.053 / 0.946		0.025	0.482		0.025	0.433

*Note: This table shows the estimates of the Community Forest Program (CFP) on Participating Chiefdoms grids, Conservation Areas, and Communal Land. The dependent variables are the annual deforestation rate relative to baseline (D^{2000}) and the year before (D^{t-1}). Columns (1) to (4) present regression results based on the unmatched sample, while columns (5) to (10) use matched samples. For each sample, the first two columns report estimates using the annual deforestation rate calculated relative to the baseline forest stock in the year 2000 (D^{2000}), whereas the latter two columns use the annual deforestation rate relative to forest stock in the previous year (D^{t-1}). Standard deviations are reported in parentheses. Errors are clustered at the chiefdom level but wild cluster bootstrapping p-values are also displayed. *,**,*** indicates 10%, 5% and 1% statistical significance levels respectively.*

Table 4: Magnitude of CFP's impact on conservation areas and close areas Annual Deforestation rate

Outcome	ATE	Control mean		Magnitude
		Pre (2001–2014)	Post (2015–2023)	(% of counterfactual)
Panel A: Conservation Area (CA)				
D^{t-1}	−0.1574	0.0142	0.1779	−88.5%
D^{2000}	−0.1412	0.0141	0.1613	−87.6%
Panel B: Close non-contracted chiefdoms (<60 km from CA boundary)				
D^{t-1}	−0.165	0.076	0.271	−60.7%
D^{2000}	−0.141	0.073	0.240	−58.7%

Note: Each panel reports estimates from my DiD specification with chiefdom and year fixed effects, estimated on the PSM-matched sample (nearest-neighbor, 1:1 without replacement, caliper = 0.0001). The treatment period is 2015–2023; the pre-treatment period is 2001–2014. The control group consists of outside-corridor cells matched on PSM scores. Magnitude is defined as the ATE estimate divided by the control group post-treatment mean; under the parallel trends assumption, this captures the share of the post-2015 deforestation surge averted by the program. D^{t-1} *denotes the annual deforestation rate relative to the lagged forest stock;* D^{2000} *denotes the rate relative to the forest stock in 2000.*

Table 5: Estimates of CFP's impact on Non-Participating Chiefdom's Annual Deforestation rate

	Raw Sample				*Matched Sample*			
	D^{2000}		D^{t-1}		D^{2000}		D^{t-1}	
Non-participating Chiefdom	-0.030		-0.043					
	(0.084)		(0.096)					
$<$ 60 km from CAs		-0.105*		-0.130*	-0.173***		-0.194***	
		(0.058)		(0.067)	(0.064)		(0.072)	
$\geqslant$ 60 km from CAs		0.012		0.006		-0.059		-0.067
		(0.110)		(0.126)		(0.114)		(0.130)
R^2	0.04	0.04	0.04	0.04	0.05	0.05	0.05	0.05
Observations	2,748,753	2,748,753	2,748,727	2,748,727	498,594	663,711	498,594	663,711
Chiefdom FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Matched Sample	No	No	No	No	Yes	Yes	Yes	Yes
# Chiefdoms	110	110	110	110	98	88	98	88
WCB p-value	0.745	0.082 / —	0.693	0.065 / —	0.023	0.681	0.022	0.682

*Note: This table shows the estimates of the Community Forest Program (CFP) on Non-Participating Chiefdoms grids, close areas ($\leqslant$ 60 km from conservation boundaries), and far areas ($>$ 60 km from conservation areas boundaries). Columns (1) to (4) present regression results based on the unmatched sample, while columns (5) to (10) use matched samples. For each sample, the first two columns report estimates using the annual deforestation rate calculated relative to the baseline forest stock in the year 2000 (D^{2000}) as outcomes, whereas the latter two columns use the annual deforestation rate relative to forest stock in the previous year (D^{t-1}). Standard deviations are reported in parentheses. Errors are clustered at the chiefdom level but wild cluster bootstrapping p-values are also displayed. *,**,*** indicates 10%, 5% and 1% statistical significance levels respectively.*

Table 6: CFP's effects across matching algorithms

	Caliper = 0.00005	Caliper = 0.0001	Caliper = 0.001	Caliper = 0.001 & k = 3	Mahalanobis
Panel A — Protected Areas					
ATT	-0.149***	-0.152***	-0.151***	-0.158***	-0.100
	(0.056)	(0.057)	(0.056)	(0.054)	(0.072)
WCB p	0.019	0.019	0.019	0.020	0.193
Obs.	331315	344172	347852	534865	189474
Clusters	66	66	66	72	47
Panel B — Close Non-Participating Chiefdoms (<60 km)					
ATT	-0.185**	-0.183**	-0.185**	-0.179**	-0.106
	(0.074)	(0.073)	(0.073)	(0.070)	(0.079)
WCB p	0.011	0.011	0.011	0.025	0.221
Obs.	334374	343896	345989	579830	195454
Clusters	90	90	90	92	65

*Note: This table shows the estimates of the CFP program on conservation areas (Panel A) and Close areas in non-participating chiefdoms (Panel B) across different matching algorithms. Columns 1 to 3 shows estimation using matched samples that found one pair for each treated grid but using different caliper thresholds. Column for use the same caliper as the main specification (0.001) but find 3 neighbors for each treated grid. Finally, column 5 shows the estimates using Mahalanobis distance algorithm to find matches. Errors are clustered at the chiefdom level but wild cluster bootstrapping p-values are also displayed. *,**,*** indicates 10%, 5% and 1% statistical significance levels respectively.*

References

Abman, Ryan, and Clark Lundberg. 2024. "Contracting, Market Access and Deforestation." *Journal of Development Economics* 168 (May): 103269. https://doi.org/10.1016/j.jdeveco.2024.103269. https://www.sciencedirect.com/science/article/pii/S030438782400018X.

Araujo, Rafael, Juliano Assunção, and Arthur Amorim Bragança. 2023. *The Effects of Transportation Infrastructure on Deforestation in the Amazon: A General Equilibrium Approach.* Policy Research Working Papers. The World Bank, April. https://doi.org/10.1596/1813-9450-10415. https://hdl.handle.net/10986/39728.

Baland, Jean-Marie, Pranab Bardhan, Sanghamitra Das, and Dilip Mookherjee. 2010. "Forests to the People: Decentralization and Forest Degradation in the Indian Himalayas." *World Development* 38, no. 11 (November): 1642–1656. https://doi.org/10.1016/j.worlddev.2010.03.007. https://www.sciencedirect.com/science/article/pii/S0305750X10000896.

Balboni, Clare, Aaron Berman, Robin Burgess, and Benjamin A. Olken. 2023. "The Economics of Tropical Deforestation." *Annual Review of Economics* 15, no. Volume 15, 2023 (September): 723–754. https://doi.org/10.1146/annurev-economics-090622-024705. https://www.annualreviews.org/content/journals/10.1146/annurev-economics-090622-024705.

Baldwin, Kate. 2013. "Why Vote with the Chief? Political Connections and Public Goods Provision in Zambia." *American Journal of Political Science* 57, no. 4 (October): 794–809. https://doi.org/10.1111/ajps.12023. https://onlinelibrary.wiley.com/doi/10.1111/ajps.12023.

———. 2015. *The Paradox of Traditional Chiefs in Democratic Africa.* Cambridge Studies in Comparative Politics. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9781316422335. https://www.cambridge.org/core/books/paradox-of-traditional-chiefs-in-democratic-africa/A0A5A0A8B513F792F501C82AF5935457.

Baragwanath, Kathryn, and Ella Bayi. 2020. "Collective Property Rights Reduce Deforestation in the Brazilian Amazon." *Proceedings of the National Academy of Sciences* 117, no. 34 (August): 20495–20502. https://doi.org/10.1073/pnas.1917874117. https://www.pnas.org/doi/10.1073/pnas.1917874117.

Bedi, Arjun S., Marcos Y. Nakaguma, Brandon J. Restrepo, and Matthias Rieger. 2021. "Particle Pollution and Cognition: Evidence from Sensitive Cognitive Tests in Brazil." *Journal of the Association of Environmental and Resource Economists* 8, no. 3 (May): 443–474. https://doi.org/10.1086/711592. https://www.journals.uchicago.edu/doi/10.1086/711592.

BenYishay, Ariel, Silke Heuser, Daniel Runfola, and Rachel Trichler. 2017. "Indigenous Land Rights and Deforestation: Evidence from the Brazilian Amazon." *Journal of Environmental Economics and Management*, Special issue on environmental economics in developing countries, 86 (November): 29–47. https://doi.org/10.1016/j.jeem.2017.07.008. https://www.sciencedirect.com/science/article/pii/S0095069617304813.

BioCarbon Partners. 2019. *Luangwa Valley Community Forests Project: Final Report.* Project Final Report. BioCarbon Partners / United States Agency for International Development (USAID).

Burgess, Robin, Matthew Hansen, Benjamin A. Olken, Peter Potapov, and Stefanie Sieber. 2012. "The Political Economy of Deforestation in the Tropics*." *The Quarterly Journal of Economics*

127, no. 4 (November): 1707–1754. https://doi.org/10.1093/qje/qjs034. https://doi.org/10.1093/qje/qjs034.

Chen, Jiafeng, and Jonathan Roth. 2024. "Logs with Zeros? Some Problems and Solutions*." *The Quarterly Journal of Economics* 139, no. 2 (May): 891–936. https://doi.org/10.1093/qje/qjad054. https://doi.org/10.1093/qje/qjad054.

Cisneros, Elías, Jan Börner, Stefano Pagiola, and Sven Wunder. 2022. "Impacts of Conservation Incentives in Protected Areas: The Case of Bolsa Floresta, Brazil." *Journal of Environmental Economics and Management* 111 (January): 102572. https://doi.org/10.1016/j.jeem.2021.102572. https://www.sciencedirect.com/science/article/pii/S0095069621001200.

Cisneros, Elías, and Krisztina Kis-Katos. 2022. *Unintended Consequences of Anti-Corruption Strategies*. SSRN Scholarly Paper. Rochester, NY, March. https://doi.org/10.2139/ssrn.3899498. https://papers.ssrn.com/abstract=3899498.

Correa, Juliano, Elías Cisneros, Jan Börner, Alexander Pfaff, Marcelo Costa, and Raoni Rajão. 2020. "Evaluating REDD+ at Subnational Level: Amazon Fund Impacts in Alta Floresta, Brazil." *Forest Policy and Economics* 116 (July): 102178. https://doi.org/10.1016/j.forpol.2020.102178. https://www.sciencedirect.com/science/article/pii/S1389934119301170.

Etchart, Nicolle, José Luis Freire, Margaret B. Holland, Kelly W. Jones, and Lisa Naughton-Treves. 2020. "What Happens When the Money Runs Out? Forest Outcomes and Equity Concerns Following Ecuador's Suspension of Conservation Payments." *World Development* 136 (December): 105124. https://doi.org/10.1016/j.worlddev.2020.105124. https://www.sciencedirect.com/science/article/pii/S0305750X20302515.

Fu, Shihe, V Brian Viard, and Peng Zhang. 2021. "Air Pollution and Manufacturing Firm Productivity: Nationwide Estimates for China." *The Economic Journal* 131, no. 640 (November): 3241–3273. https://doi.org/10.1093/ej/ueab033. https://doi.org/10.1093/ej/ueab033.

Giudice, Renzo, Jan Börner, Sven Wunder, and Elias Cisneros. 2019. "Selection Biases and Spillovers from Collective Conservation Incentives in the Peruvian Amazon." *Environmental Research Letters* 14, no. 4 (April): 045004. https://doi.org/10.1088/1748-9326/aafc83. https://dx.doi.org/10.1088/1748-9326/aafc83.

Global Forest Watch. 2026. *Global Forest Watch Interactive Map*. https://www.globalforestwatch.org.

Gollin, Douglas, and Julien Wolfersberger. n.d. "Agricultural Trade and Deforestation: The Role of New Roads."

Gulzar, Saad, Apoorva Lal, and Benjamin Pasquale. 2024. "Representation and Forest Conservation: Evidence from India's Scheduled Areas." *American Political Science Review* 118, no. 2 (May): 764–783. https://doi.org/10.1017/S0003055423000758. https://www.cambridge.org/core/journals/american-political-science-review/article/representation-and-forest-conservation-evidence-from-indias-scheduled-areas/BF8BB4821C97E6E67E682D71345DA0FE.

Hansen, James, Makiko Sato, Gary Russell, and Pushker Kharecha. 2013. "Climate Sensitivity, Sea Level and Atmospheric Carbon Dioxide." *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 371, no. 2001 (October): 20120294. https://doi.org/10.1098/rsta.2012.0294. https://royalsocietypublishing.org/doi/10.1098/rsta.2012.0294.

Heyes, Anthony, and Soodeh Saberian. 2024. "Pollution and Learning: Causal Evidence from Obama's Iran Sanctions." *Journal of Environmental Economics and Management* 125 (May): 102965. https://doi.org/10.1016/j.jeem.2024.102965. https://linkinghub.elsevier.com/retrieve/pii/S0095069624000391.

Huntington, Heather, Aleta Haflett Starosta, Stephanie Fenner, and Ben Ewing. 2016. *Report on Baseline Findings: Community Forest Management Program.* Impact Evaluation Baseline Report. The Cloudburst Group, December.

Izquierdo-Tort, Santiago, Seema Jayachandran, and Santiago Saavedra. 2024. "Redesigning Payments for Ecosystem Services to Increase Cost-Effectiveness." *Nature Communications* 15, no. 1 (October): 9252. https://doi.org/10.1038/s41467-024-53643-1. https://www.nature.com/articles/s41467-024-53643-1.

Jack, B. Kelsey, Seema Jayachandran, Namrata Kala, and Rohini Pande. 2022. *Money (Not) to Burn: Payments for Ecosystem Services to Reduce Crop Residue Burning.* Working Paper, November. https://doi.org/10.3386/w30690. https://www.nber.org/papers/w30690.

Jayachandran, Seema. 2022. *The Inherent Trade-Off Between the Environmental and Anti-Poverty Goals of Payments for Ecosystem Services.* Working Paper, April. https://doi.org/10.3386/w29954. https://www.nber.org/papers/w29954.

Jayachandran, Seema, Joost de Laat, Eric F. Lambin, Charlotte Y. Stanton, Robin Audy, and Nancy E. Thomas. 2017. "Cash for Carbon: A Randomized Trial of Payments for Ecosystem Services to Reduce Deforestation." *Science* 357, no. 6348 (July): 267–273. https://doi.org/10.1126/science.aan0568. https://www.science.org/doi/10.1126/science.aan0568.

Leventon, Julia, Felix K. Kalaba, Jen C. Dyer, Lindsay C. Stringer, and Andrew J. Dougill. 2014. "Delivering Community Benefits Through REDD+: Lessons from Joint Forest Management in Zambia." *Forest Policy and Economics* 44 (July): 10–17. https://doi.org/10.1016/j.forpol.2014.03.005. https://linkinghub.elsevier.com/retrieve/pii/S1389934114000549.

Lindsey, Peter A., Vincent R. Nyirenda, Jonathan I. Barnes, Matthew S. Becker, Rachel McRobb, Craig J. Tambling, W. Andrew Taylor, et al. 2014. "Underperformance of African Protected Area Networks and the Case for New Conservation Models: Insights from Zambia." *PLOS ONE* 9, no. 5 (May): e94109. https://doi.org/10.1371/journal.pone.0094109. https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0094109.

Malan, Mandy, Rachel Carmenta, Elisabeth Gsottbauer, Paul Hofman, Andreas Kontoleon, Tom Swinfield, and Maarten Voors. 2024. "Evaluating the Impacts of a Large-Scale Voluntary REDD+ Project in Sierra Leone." *Nature Sustainability* 7, no. 2 (February): 120–129. https://doi.org/10.1038/s41893-023-01256-9. https://www.nature.com/articles/s41893-023-01256-9.

Manda, Simon, and Nyambe Mukanda. 2023. "Can REDD+ Projects Deliver Livelihood Benefits in Private Tenure Arrangements? Experiences from Rural Zambia." *Forest Policy and Economics* 150 (May): 102952. https://doi.org/10.1016/j.forpol.2023.102952. https://www.sciencedirect.com/science/article/pii/S1389934123000473.

Nash, J, N Halewood, and S Melhem. 2013. "Unlocking Africa's Agricultural Potential: An Action Agenda for Transformation'Africa Region Sustainable Development Series." *World Bank Group, Washington DC*, 1–70.

Odo, Daniel B., Ian A. Yang, Sagnik Dey, Melanie S. Hammer, Aaron Van Donkelaar, Randall V. Martin, Guang-Hui Dong, et al. 2023. "A Cross-Sectional Analysis of Long-Term Exposure

to Ambient Air Pollution and Cognitive Development in Children Aged 3–4 Years Living in 12 Low- and Middle-Income Countries." *Environmental Pollution* 318 (February): 120916. https://doi.org/10.1016/j.envpol.2022.120916. https://linkinghub.elsevier.com/retrieve/pii/S0269749122021315.

Rambachan, Ashesh, and Jonathan Roth. 2023. "A More Credible Approach to Parallel Trends." *The Review of Economic Studies* 90, no. 5 (October): 2555–2591. https://doi.org/10.1093/restud/rdad018. https://doi.org/10.1093/restud/rdad018.

Rico-Straffon, Jimena, Zhenhua Wang, Stephanie Panlasigui, Colby J. Loucks, Jennifer Swenson, and Alexander Pfaff. 2023. "Forest Concessions and Eco-Certifications in the Peruvian Amazon: Deforestation Impacts of Logging Rights and Logging Restrictions." *Journal of Environmental Economics and Management* 118 (March): 102780. https://doi.org/10.1016/j.jeem.2022.102780. https://www.sciencedirect.com/science/article/pii/S0095069622001334.

Roth, Jonathan. 2022. "Pretest with Caution: Event-Study Estimates After Testing for Parallel Trends." *American Economic Review: Insights* 4, no. 3 (September): 305–22. https://doi.org/10.1257/aeri.20210236. https://www.aeaweb.org/articles/pdf/doi/10.1257/aeri.20210236.

Sims, Katharine R. E. 2010. "Conservation and Development: Evidence from Thai Protected Areas." *Journal of Environmental Economics and Management* 60, no. 2 (September): 94–114. https://doi.org/10.1016/j.jeem.2010.05.003. https://www.sciencedirect.com/science/article/pii/S0095069610000586.

Sze, Jocelyne S., L. Roman Carrasco, Dylan Childs, and David P. Edwards. 2022. "Reduced Deforestation and Degradation in Indigenous Lands Pan-Tropically." *Nature Sustainability* 5, no. 2 (February): 123–130. https://doi.org/10.1038/s41893-021-00815-2. https://www.nature.com/articles/s41893-021-00815-2.

ONLINE APPENDIX
A Strategic Environment: Community Conservation, Site Selection, and Spillovers in Eastern Zambia

A Additional Institutional Details

A.1 Zambia conservation policies

The CFP was implemented within a rapidly changing national conservation policy environment. Community-based natural resource management (CBNRM) in Zambia traces its origins to the Administrative Management Design for Game Management Areas (ADMADE) program, launched in the Luangwa Valley in the late 1980s, which first established community resource boards (CRBs) and community scouts as the institutional backbone of wildlife governance. The Wildlife Act No. 14 of 2015 restructured this framework substantially: it dissolved the semi-autonomous Zambia Wildlife Authority (ZAWA) and replaced it with the Department of National Parks and Wildlife (DNPW) under direct government control, re-centralizing institutional authority at precisely the moment the CFP was attempting to devolve conservation rights to communities Lindsey et al. 2014. Compounding this, a presidential moratorium on trophy hunting in Game Management Areas, declared in 2013 and maintained in the 2015 Act, eliminated the primary revenue stream that had historically funded CRBs through safari licensing fees. Enacted in the same year, the Forests Act No. 4 of 2015 introduced provisions for joint and community forest management but did not establish statutory community rights to carbon credits — meaning carbon generated on customary land remained legally state property. The CFP therefore operated in a legal grey zone, transferring carbon rights to communities through private voluntary contracts in the absence of any national statutory framework (Manda and Mukanda 2023; Leventon et al. 2014). This context helps explain both community willingness to sign long-term conservation contracts — given the collapse of alternative CRB revenues — and the program's reliance on the voluntary carbon market rather than government-mediated REDD+ payments.

B Additional Results (Online)

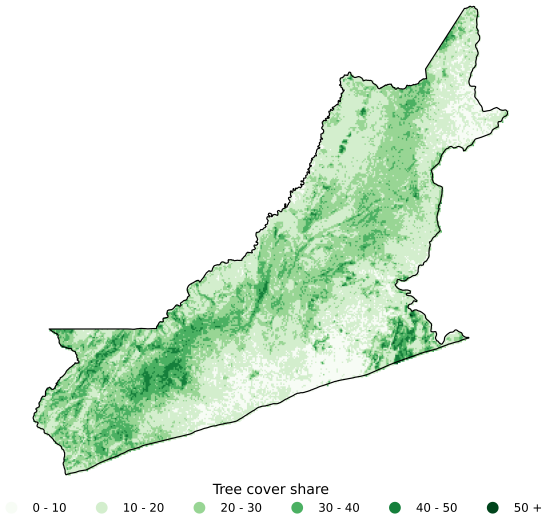
B.1 Sensibility and robustness analysis

To test the sensitivity of my estimates to the parallel trends assumption, I follow Rambachan and Roth (2023) and estimate robust confidence intervals under the Relative Magnitudes (RM) restriction, which bounds post-treatment violations of parallel trends by $\bar{M}$ times the maximum pre-treatment deviation. Figure D.4 shows the results for conservation areas (D.4a) and close areas (D.4b). The robust confidence interval excludes zero at $\bar{M} = 0$ but includes zero at $\bar{M} \approx 0.5$, indicating that the results are sensitive to post-treatment violations exceeding 50% of the largest pre-period deviation. This sensitivity is consistent with the documented selection bias in conservation site placement, which creates pre-existing differences in deforestation trends between treated and control grids.

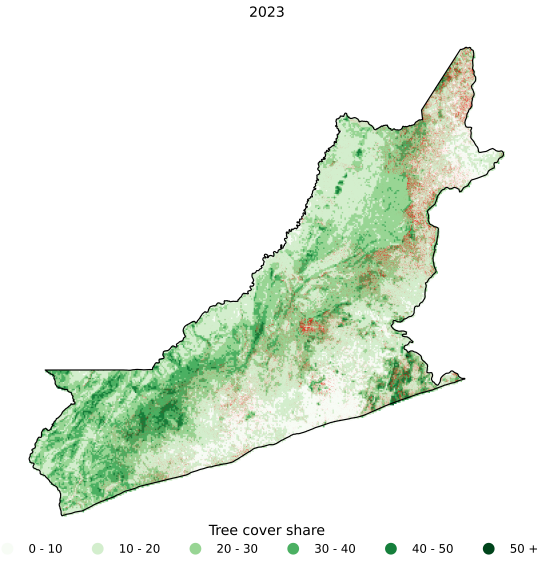
I also assess the power of the pre-trends test following Roth (2022) an plot these results in figure D.5). Results indicate that the joint pre-trends test has very low power: the minimum detectable slope at 50% power is approximately 0.001 percentage points per year for conservation areas — economically negligible relative to the estimated ATE of -0.14 pp. Passing the pre-trends test therefore provides limited assurance of parallel trends. However, the hypothesized linear violation predicted by the test is slightly positive, while my post-treatment coefficients are strongly negative (reaching -0.39 pp by 2023). Since the hypothesized confounding trend and the estimated effects run in opposite directions, a linear pre-existing trend cannot account for the post-treatment pattern, providing directional evidence in favor of a causal interpretation.

C Appendix Figures and Table (Online)

Fig. D.1. Eastern Province baseline tree cover and tree cover loss in 2023



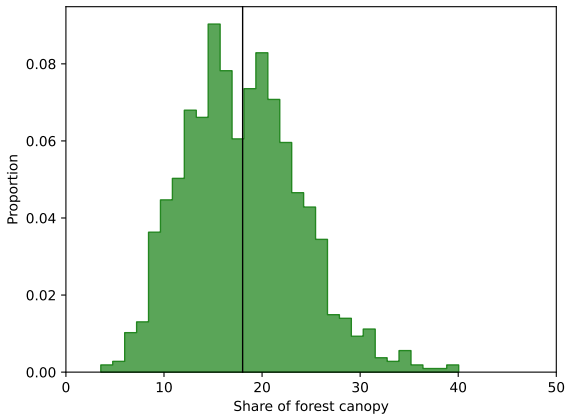
(a) Tree cover in 2000



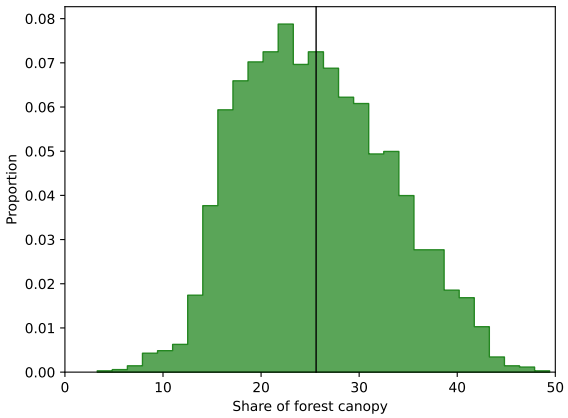
(b) Cell tree cover loss in 2023

Note: The maps plot tree cover in 2000 and tree cover loss in 2023, using 30m × 30m Hansen cells. Figure D.1a maps the tree cover in 2000 for each cell, and Figure D.1b highlights the deforested cells in 2023, shown in red.

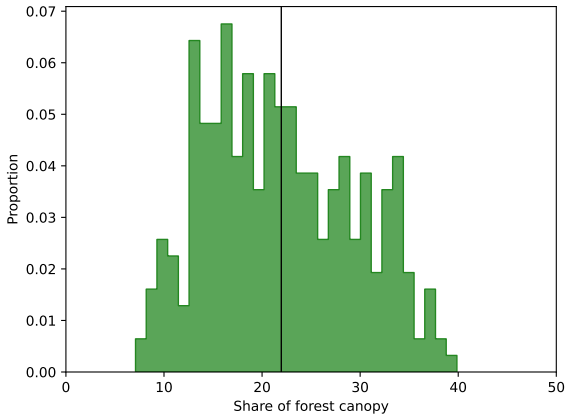
Fig. D.2. Chiefdom distribution of 0.1-degree cell share of tree cover



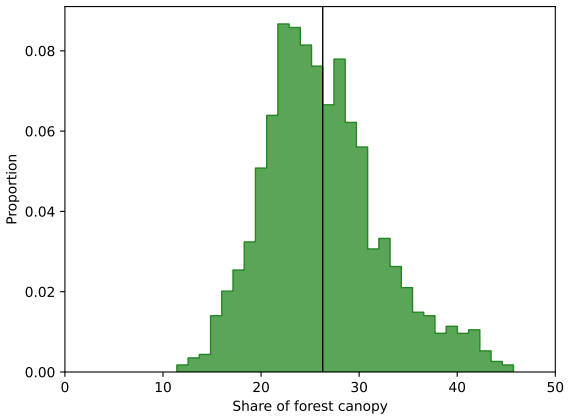
(a) Chikomene



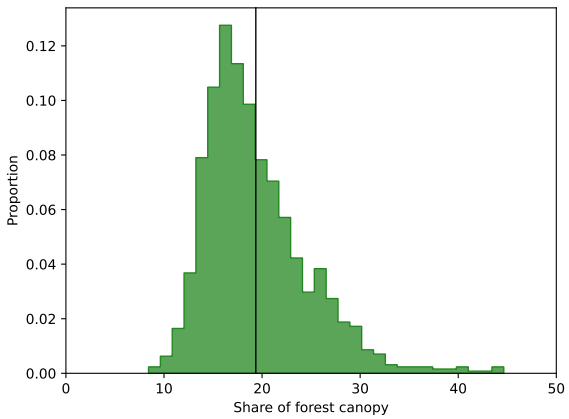
(b) Luembe



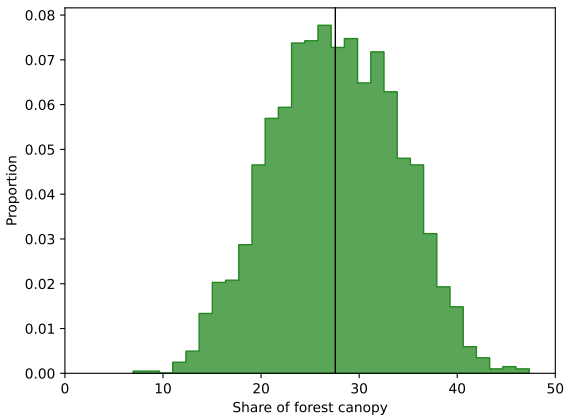
(c) Malama



(d) Msoro



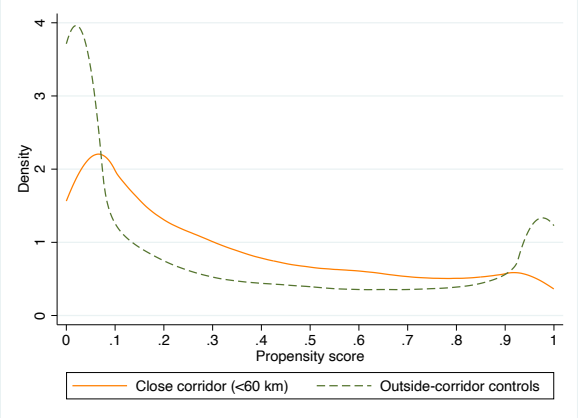
(e) Mwanya



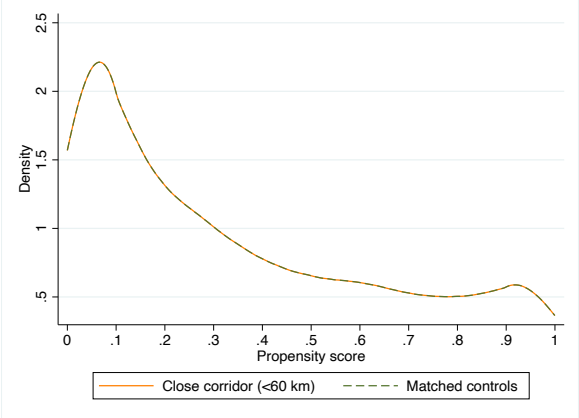
(f) Nyalugwe

Note: This Figure presents the 0.1 degree cell share of tree cover distributions of Chiefdoms surveyed by USAID in 2015 and 2024. The x-axis corresponds to the cell share of forest canopy, i.e., the proportion of the cell which is populated by crowns of trees. The y-axis corresponds to the proportion of cells within a share of canopy bin.

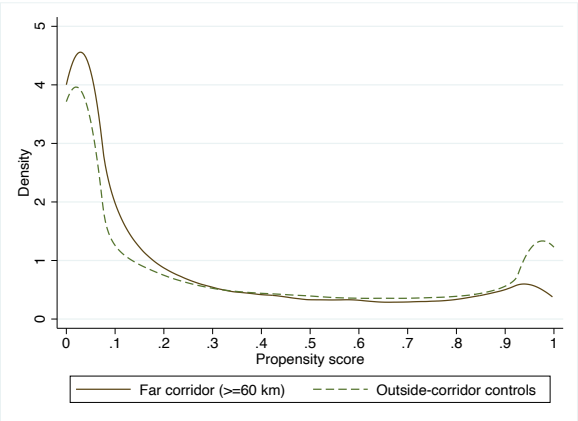
Fig. D.3. Grid propensity score distributions for non-participating chiefdoms



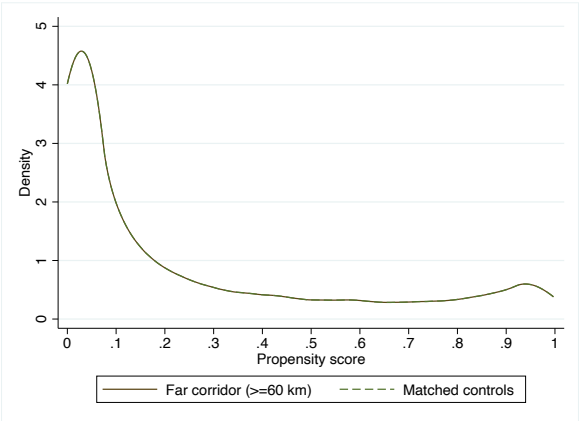
(a) Close: Before



(b) Close: After



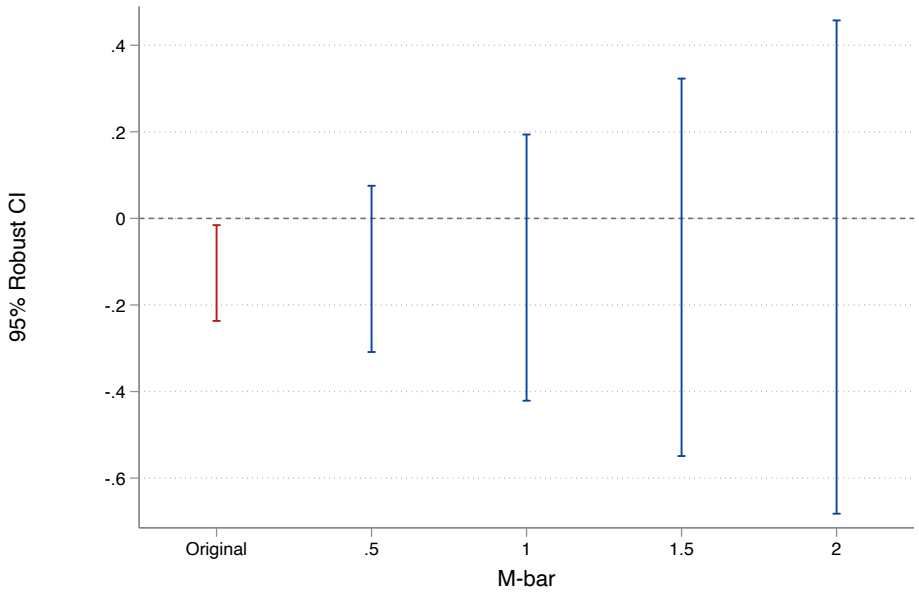
(c) Far: Before



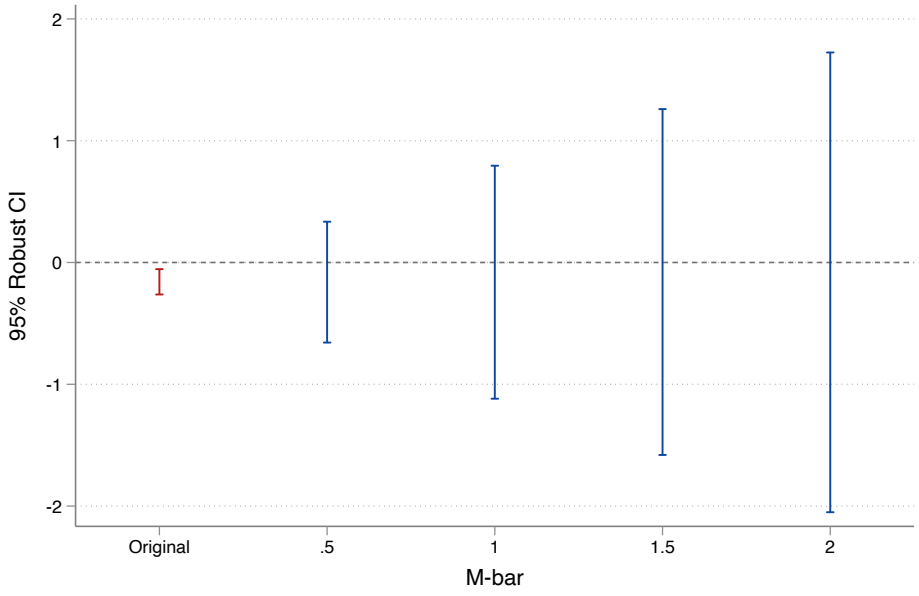
(d) Far: After

Note: Panels D.3a and D.3b show propensity score distributions for close non-participating cells ($\leqslant$ 60 km from Conservation Area boundaries) and their matched controls before and after matching. Panels D.3c and D.3d show the same for far non-participating cells ($\geqslant$ 60 km). Propensity scores are estimated using equation 4.1.

Fig. D.4. Sensitivity analysis for conservation and close areas ATE effects



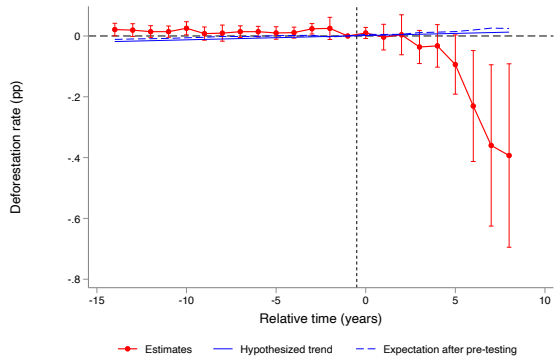
(a) Conservation areas



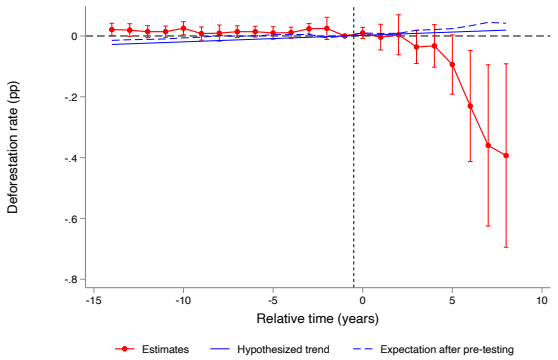
(b) Close areas

Note: This figure reports sensitivity analysis results following Rambachan and Roth (2023), . implemented via the `honestdid` package. Each panel plots the robust 95% confidence interval for the average treatment effect (ATE) as a function of $\bar{M}$, the maximum allowed ratio of post-treatment to pre-treatment violations of parallel trends under the Relative Magnitudes (RM) restriction. The ATE is the equal-weighted average treatment effect over the 9 post-treatment periods (2015–2023). Panel D.4a shows results for Conservation Areas. Panel D.4b shows results for close non-contracted chiefdoms. Estimation uses the PSM-matched sample with chiefdom and year fixed effects and errors clustered at the chiefdom level.

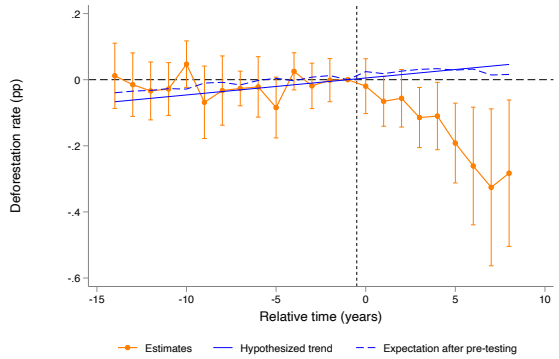
Fig. D.5. Pretrends tests for conservation and close areas event study estimates



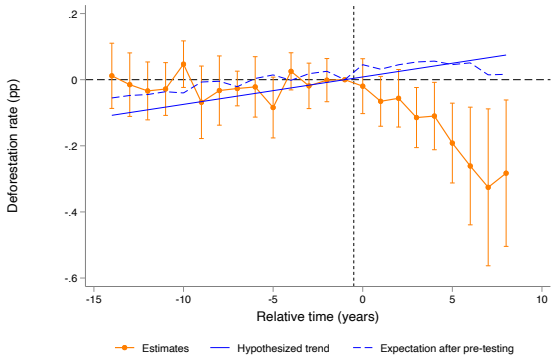
(a) Conservation areas : Slope 50%



(b) Conservation areas : Slope 80%



(c) Close areas : Slope 50%



(d) Close areas : Slope 80 %

Note: This figure reports pre-trends power calculations following Roth (2022), implemented via the `pretrends` package. Each panel plots three elements on the event-study plot: (i) the estimated event-study coefficients with 95% confidence intervals (colored markers and caps); (ii) the hypothesized linear trend (blue solid line); and (iii) the expected event-study path conditional on passing the pre-trends test under the hypothesized violation (blue dashed line). The x-axis shows years relative to program start; the reference year is 2014. Panels D.5a, D.5b report results for Conservation Areas at 50% and 80% detection power, respectively. Panels D.5c, and D.5d report results for Close Areas.

Table D.1: Pre-trend slope estimates for Conservation Areas and Communal Land (D^{t-1})

	CA unmatched	CA matched	CL unmatched	CL matched
Pre-trend slope	-0.004	-0.000	-0.003	0.001
	(0.003)	(0.001)	(0.003)	(0.003)
R^2	0.24	0.13	0.24	0.19
N	1,435,784	195,202	1,435,784	208,796
Outcome	D^{t-1}	D^{t-1}	D^{t-1}	D^{t-1}
Pre-period	2001-2014	2001-2014	2001-2014	2001-2014
Cell FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Note: This table reports pre-trend slope estimates from a linear regression of event-study coefficients on relative time, restricted to the pre-treatment period (2001–2014). A slope close to zero indicates that treated and control cells followed parallel trends before the program. Columns compare unmatched and matched samples for Conservation Areas (CA) and Communal Land (CL) within contracted chiefdoms. The dependent variable is D^{t-1}*. Matching uses nearest-neighbor 1:1 without replacement, caliper = 0.0001. Standard errors clustered at the chiefdom level are in parentheses; wild cluster bootstrap* p*-values are also displayed. *, **, *** indicate 10%, 5%, and 1% significance levels, respectively.*

Table D.2: Pre-trend slope estimates for Conservation Areas and Communal Land (D^{2000})

	CA unmatched	CA matched	CL unmatched	CL matched
Pre-trend slope	-0.003	-0.000	-0.002	0.001
	(0.003)	(0.001)	(0.003)	(0.003)
R^2	0.22	0.12	0.22	0.18
N	1,435,784	195,202	1,435,784	208,796
Outcome	D^{2000}	D^{2000}	D^{2000}	D^{2000}
Pre-period	2001-2014	2001-2014	2001-2014	2001-2014
Cell FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Note: This table reports pre-trend slope estimates from a linear regression of event-study coefficients on relative time, restricted to the pre-treatment period (2001–2014). A slope close to zero indicates that treated and control cells followed parallel trends before the program. Columns compare unmatched and matched samples for Conservation Areas (CA) and Communal Land (CL) within contracted chiefdoms. The dependent variable is D^{2000}*. Matching uses nearest-neighbor 1:1 without replacement, caliper = 0.0001. Standard errors clustered at the chiefdom level are in parentheses; wild cluster bootstrap* p*-values are also displayed. *, **, *** indicate 10%, 5%, and 1% significance levels, respectively.*

Table D.3: Heterogeneous estimates of CFP's impact on conservation areas' annual deforestation rate

	D^{2000}				D^{t-1}			
CA	-0.395**	-0.196***	-0.265**	-0.115	-0.438**	-0.221***	-0.303**	-0.127
	(0.176)	(0.072)	(0.122)	(0.083)	(0.196)	(0.081)	(0.139)	(0.094)
CA x Distance to CA border (km)	-0.003				-0.004			
	(0.003)				(0.003)			
CA x Distance to roads (km)		0.006*				0.006*		
		(0.003)				(0.004)		
CA x Distance to Settlements (km)			0.018				0.022*	
			(0.011)				(0.013)	
CA x Tree cover in 2000				-0.001				-0.001
				(0.002)				(0.002)
R^2	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04
Observations	320,689	320,689	320,689	320,689	320,689	320,689	320,689	320,689
Chiefdom FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Matched Sample	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

*Note: This table shows the estimates of the heterogeneous effects of the Community Forest Program (CFP) on Conservation areas grids. The dependent variables are the annual deforestation rate relative to baseline (D^{2000}) and the year before (D^{t-1}). Each column display the estimates for a different regression containing the interaction between conservation area grids status and grid characteristics. All columns shows results using the matched sample using the main matching algorithm. Standard deviations are reported in parentheses. Errors are clustered at the chiefdom level but wild cluster bootstrapping p-values are also displayed. *,**,*** indicates 10%, 5% and 1% statistical significance levels respectively.*

Table D.4: Heterogeneous estimates of CFP's impact on close areas' annual deforestation rate

	D^{2000}				D^{t-1}			
Close	-0.454***	-0.234***	-0.257**	-0.028	-0.511***	-0.264***	-0.293**	-0.027
	(0.133)	(0.087)	(0.108)	(0.068)	(0.150)	(0.100)	(0.124)	(0.076)
Close x Distance to PA border (km)	0.001				0.001			
	(0.001)				(0.001)			
Close x Distance to roads (km)		0.008				0.009		
		(0.005)				(0.006)		
Close x Distance to Settlements (km)			0.021				0.024	
			(0.013)				(0.015)	
Close x Tree cover in 2000				-0.005				-0.006
				(0.003)				(0.004)
R^2	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
Observations	498,594	498,594	498,594	498,594	498,594	498,594	498,594	498,594
Chiefdom FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Matched Sample	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

*Note: This table shows the estimates of the heterogeneous effects of the Community Forest Program (CFP) on close grids ($\leqslant$ 60 km from conservation areas boundaries). The dependent variables are the annual deforestation rate relative to baseline (D^{2000}) and the year before (D^{t-1}). Each column display the estimates for a different regression containing the interaction between conservation area grids status and grid characteristics. All columns shows results using the matched sample using the main matching algorithm. Standard deviations are reported in parentheses. Errors are clustered at the chiefdom level but wild cluster bootstrapping p-values are also displayed. *,**,*** indicates 10%, 5% and 1% statistical significance levels respectively.*

Table D.5: Spillover effects by distance from Conservation Area boundaries (D^{2000})

	CL Unmatched	CL Matched	Close Unmatched	Close Matched
CL 0–10 km	-0.037	-0.116		
	(0.065)	(0.076)		
CL 10–20 km	0.091	0.012		
	(0.088)	(0.096)		
CL 20–30 km	0.412*	0.333		
	(0.215)	(0.220)		
CL > 30 km	0.024	-0.056		
	(0.057)	(0.070)		
0–15 km			-0.129**	-0.165***
			(0.057)	(0.059)
15–30 km			-0.095	-0.130**
			(0.061)	(0.063)
30–45 km			-0.088	-0.124*
			(0.062)	(0.064)
45–60 km			-0.096	-0.132
			(0.081)	(0.083)
R^2	0.04	0.05	0.04	0.05
N	2,188,956	343,022	2,178,675	323,702
Chiefdom FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
PSM matched	No	Yes	No	Yes
Reference	Controls	Controls	Controls	Controls
— CL WCB p-values —				
CL 0–10 km	0.193	0.011		
CL 10–20 km	0.447	0.618		
CL 20–30 km	0.065	0.056		
CL > 30 km	0.304	0.290		
— Close WCB p-values —				
Close 0–15 km			0.073	0.018
Close 15–30 km			0.131	0.091
Close 30–45 km			0.141	0.059
Close 45–60 km			0.423	0.587
Chiefdom FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
PSM matched	No	Yes	No	Yes
Reference	Controls	Controls	Controls	Controls

Note:This table presents spillover effects of the CFP at different distances from Conservation Area boundaries, for Communal Land within contracted chiefdoms (left panels) and for close non-contracted chiefdoms (right panels). Distance bands for Communal Land are 0–10 km, 10–20 km, 20–30 km, and >30 km. Distance bands for close areas are 0–15 km, 15–30 km, 30–45 km, and 45–60 km. D^{2000} *is the outcome deforestation rate reported. A declining gradient in absolute effect size with distance is consistent with monitoring or knowledge spillovers operating locally around Conservation Area boundaries. All estimates use the matched sample. Standard errors clustered at the chiefdom level are in parentheses; wild cluster bootstrap* p*-values are also displayed. *, **, *** indicate 10%, 5%, and 1% significance levels, respectively.*

Table D.6: Spillover effects by distance from Conservation Area boundaries (D^{t-1}

	CL Unmatched	CL Matched	Close Unmatched	Close Matched
CL 0–10 km	-0.063	-0.141		
	(0.074)	(0.085)		
CL 10–20 km	0.076	-0.003		
	(0.097)	(0.107)		
CL 20–30 km	0.469*	0.392		
	(0.278)	(0.283)		
CL > 30 km	0.011	-0.068		
	(0.067)	(0.080)		
0–15 km			-0.159**	-0.192***
			(0.066)	(0.068)
15–30 km			-0.123*	-0.155**
			(0.070)	(0.072)
30–45 km			-0.113	-0.145*
			(0.072)	(0.074)
45–60 km			-0.120	-0.152
			(0.091)	(0.093)
R^2	0.04	0.06	0.04	0.05
N	2,188,956	343,022	2,178,675	323,702
Chiefdom FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
PSM matched	No	Yes	No	Yes
Reference	Controls	Controls	Controls	Controls
— CL WCB p-values —				
CL 0–10 km	0.118	0.011		
CL 10–20 km	0.529	0.683		
CL 20–30 km	0.066	0.057		
CL > 30 km	0.319	0.297		
— Close WCB p-values —				
Close 0–15 km			0.062	0.017
Close 15–30 km			0.100	0.080
Close 30–45 km			0.126	0.056
Close 45–60 km			0.343	0.558
Chiefdom FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
PSM matched	No	Yes	No	Yes
Reference	Controls	Controls	Controls	Controls

Note: This table presents spillover effects of the CFP at different distances from Conservation Area boundaries, for Communal Land within contracted chiefdoms (left panels) and for close non-contracted chiefdoms (right panels). Distance bands for Communal Land are 0–10 km, 10–20 km, 20–30 km, and >30 km. Distance bands for close areas are 0–15 km, 15–30 km, 30–45 km, and 45–60 km. D^{t-1} *is the outcome deforestation rate reported. A declining gradient in absolute effect size with distance is consistent with monitoring or knowledge spillovers operating locally around Conservation Area boundaries. All estimates use the matched sample. Standard errors clustered at the chiefdom level are in parentheses; wild cluster bootstrap* p*-values are also displayed. *, **, *** indicate 10%, 5%, and 1% significance levels, respectively.*

Table D.7: Robustness of CFP effects by time chunk inclusion

	D^{2000}				D^{T-1}			
	Main	PES full	PES no pilot	PES 2019-21	Main	PES full	PES no pilot	PES 2019-21
Conservation Area	-0.141**	-0.232***	-0.237***	-0.129**	-0.157**	-0.259***	-0.264***	-0.142**
	(0.055)	(0.085)	(0.088)	(0.052)	(0.062)	(0.095)	(0.099)	(0.058)
R^2	0.12	0.13	0.14	0.12	0.13	0.14	0.15	0.13
Observations	320,689	320,689	264,917	292,803	320,689	320,689	264,917	292,803
After = 2015+	Yes	No	No	No	Yes	No	No	No
After = 2019+	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Cell FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
PSM matched	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pilot yrs (15-18)	Yes	Yes	No	Yes	Yes	Yes	No	Yes
Drop 2022-23	No	No	No	Yes	No	No	No	Yes
WCB p-value (CA)	0.025	0.012	0.018	0.021	0.025	0.014	0.016	0.022

Note: This table shows the robustness of CFP estimates for Conservation Areas across alternative treatment window definitions. Column (1) uses the full post-2015 period. Column (2) shifts the treatment year to 2019, coinciding with the start of PES carbon market payments under Verra VCS certification. Column (3) drops the pilot period (2015–2018). Column (4) drops the post-2019 period. The larger negative effects in column (2) are consistent with the two-phase program design: flat-fee USAID payments (2015–2018) provided weaker conservation incentives than the performance-based PES carbon revenues introduced in 2019. All estimates use the matched sample and both D^{2000} and D^{t-1} outcomes. Standard errors clustered at the chiefdom level are in parentheses; wild cluster bootstrap p-values are also displayed. *, **, *** indicate 10%, 5%, and 1% significance levels, respectively.

Table D.8: Important terminologies

Terminology	Acronym	Description
Chiefdom	—	Traditional administrative subdivision of a district in Zambia, governed by a hereditary chief. The program's contracting unit.
Game Management Area	GMA	State-designated buffer zone surrounding a National Park where regulated hunting and community use are permitted.
National Park	NP	Strictly protected area under Zambian Wildlife Authority (ZAWA) jurisdiction; no consumptive use allowed.
Participating	—	Chiefdoms that signed a formal conservation contract with the USAID program (Luangwa Valley CFP), committing to forest protection in exchange for payments and development support.
Non-participating	—	Chiefdoms adjacent to the program corridor that did not sign a contract and received no direct program benefits. Used as a control group.
Conservation Area	CA	Area within a participating chiefdom formally set aside for wildlife conservation under the contract. Direct treatment unit.
Communal Land	CL	Remaining land within a participating chiefdom outside the Conservation Area boundary, where communities retain use rights.
Close areas	—	Grid cells in non-participating chiefdoms located within 60 km of the nearest Conservation Area boundary. Used to assess spillover effects.

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Table D.8 *(continued)*

Group	Acronym	Description
Far areas	—	Grid cells in non-participating chiefdoms located more than 60 km from the nearest Conservation Area boundary. Used as a baseline control subsample.